

Lec. 1

Symmetric Ciphers

Cryptography is probably the most important aspect of communications security and is becoming increasingly important as a basic building block for computer security.

The increased use of computer and communications systems by industry has increased the risk of theft of proprietary information. Although these threats may require a variety of countermeasures, encryption is a primary method of protecting valuable electronic information.

By far the most important automated tool for network and communications security is encryption. Two forms of encryption are in common use: conventional, or symmetric, encryption and public-key, or asymmetric, encryption.

Part One provides a survey of the basic principles of symmetric encryption, looks at widely used algorithms, and discusses applications of symmetric cryptography.

Cryptology

Is the science and study of systems for secret communications. It consists of two complementary fields of study: **Cryptography**, the design of secret communications systems, and **Cryptanalysis**, the study of ways to compromise of secret communications systems.

Cryptology primarily has been applied in military and diplomatic communications systems, but other significant applications are becoming apparent.

Cryptography methods applied by authorized information sharers to design and develop encryption schemes in order to ensure confidentiality of information.

Crypt-Analysis (mathematical and statistical attempts by unauthorized persons to break cipher in order to reveal the meaning of the underlying protected data).

Cryptography

Is the study of mathematical techniques related to aspects of information security such as confidentiality, data integrity, entity authentication, and data origin authentication.

Cryptography is not the only means of providing information security, but rather one set of techniques. Cryptography classified into two type:

1- Symmetric

2- Asymmetric

as shown in Fig. (2.1).

A Symmetric cryptography ciphers may in fact be sub classified into **Block Ciphers** (in which blocks of data, known as plaintext, are transformed into cipher text which appears unintelligible to unauthorized persons) and **Stream Ciphers** (which involve streams of typically binary operations and are well suited for efficient computer implementation).

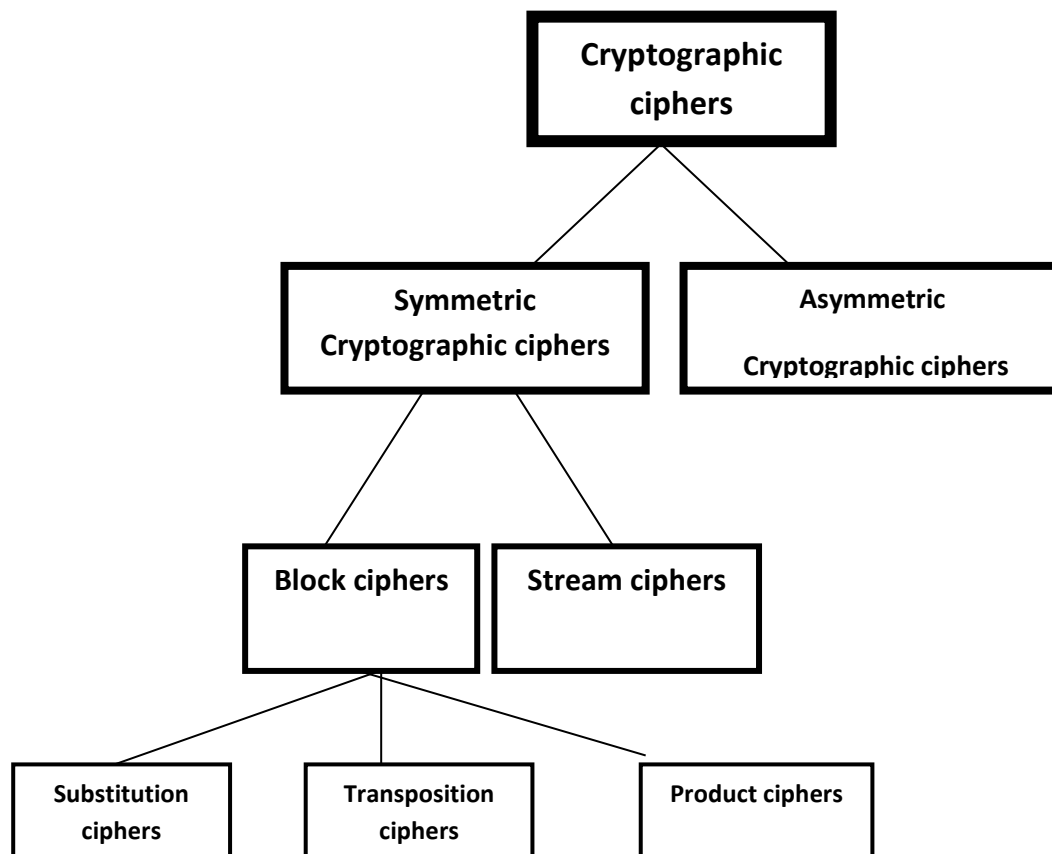


Figure 2.1 : Schematic representation of cryptographic cipher classification

The process of transforming plaintext into ciphertext is called **Encryption**; the reverse process of transforming ciphertext into plaintext is called **Decryption**. Both encryption and decryption are controlled by cryptographic key parameters as shown in Fig 3.2.

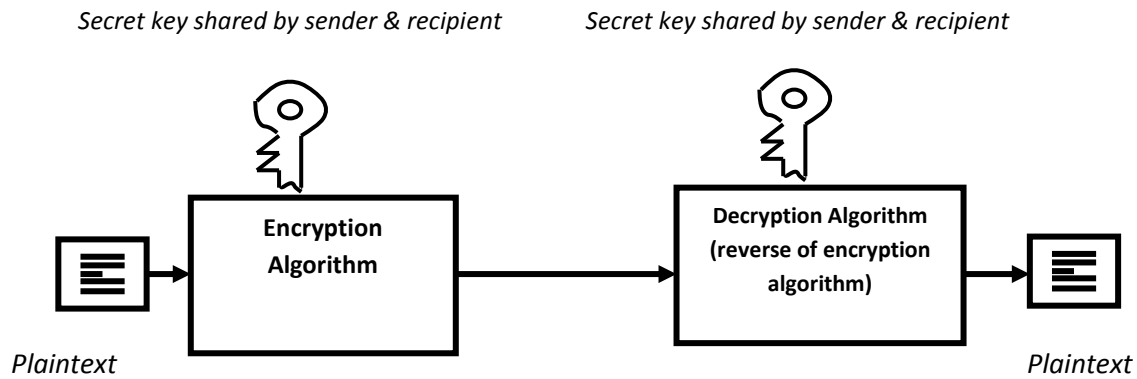
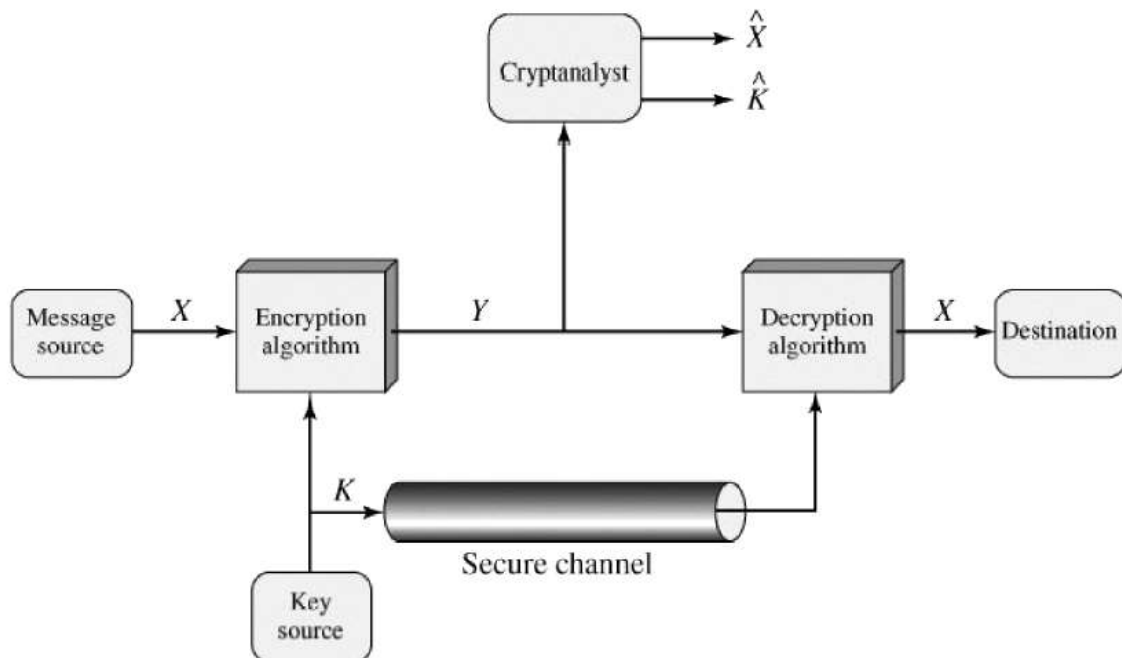


Figure 3.2 : Simplified Model of Conventional Encryption

Model for Network Security

A model for much of what we will be discussing is captured, in very general terms, in [Figure 1.5](#). A message is to be transferred from one party to another across some sort of internet. The two parties, who are the *principals* in this transaction, must cooperate for the exchange to take place. A logical information channel is established by defining a route through the internet from source to destination and by the cooperative use of communication protocols (e.g., TCP/IP) by the two principals.



Security aspects come into play when it is necessary or desirable to protect the information transmission from an opponent who may present a threat to confidentiality, authenticity, and so on. All the techniques for providing security have two components:

A security-related transformation on the information to be sent. Examples include the encryption of the message, which scrambles the message so that it is unreadable by the opponent, and the addition of a code based on the contents of the message, which can be used to verify the identity of the sender. Some secret information shared by the two principals and, it is hoped, unknown to the o. An example is an encryption key used in conjunction with the transformation to scramble the message before transmission and unscramble it on reception.

Part Two discusses a form of encryption, known as public-key encryption, in which only one of the two principals needs to have the secret information.

Secret information to the two principals while keeping it from any opponent. Or a third party may be needed to arbitrate disputes between the two principals concerning the authenticity of a message transmission.

This general model shows that there are four basic tasks in designing a particular security service:

1. Design an algorithm for performing the security-related transformation. The algorithm should be such that an opponent cannot defeat its purpose.
2. Generate the secret information to be used with the algorithm.
3. Develop methods for the distribution and sharing of the secret information.
4. Specify a protocol to be used by the two principals that makes use of the security algorithm and the secret information to achieve a particular security service.

Parts One through Three of this book concentrates on the types of security mechanisms and services that fit into the model shown in Figure 1.5. However, there are other security-related situations of interest that do not neatly fit this model but that are considered in this book. A general model of these other situations is illustrated by Figure 1.6, which reflects a concern for protecting an information system from unwanted access. Most readers are familiar with the concerns caused by the existence of hackers, who attempt to penetrate systems that can be accessed over a network. The hacker can be someone who, with no malign intent, simply gets satisfaction from breaking and entering a computer system. Or, the intruder can be a disgruntled employee who wishes to do damage, or a criminal who seeks to exploit computer assets for financial gain (e.g., obtaining credit card numbers or performing illegal money transfers).

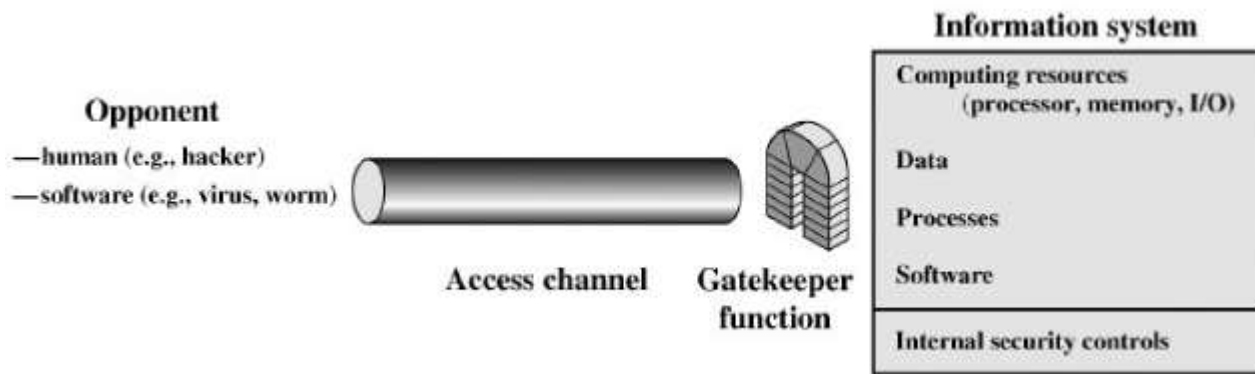


Figure 1.6. Network Access Security Model

Another type of unwanted access is the placement in a computer system of logic that exploits vulnerabilities in the system and that can affect application programs as well as utility programs, such as editors and compilers. Programs can present two kinds of threats:

Information access threats intercept or modify data on behalf of users who should not have access to that data.

Service threats exploit service flaws in computers to inhibit use by legitimate users.

Viruses and worms are two examples of software attacks. Such attacks can be introduced into a system by means of a disk that contains the unwanted logic concealed in otherwise useful software. They can also be inserted into a system across a network; this latter mechanism is of more concern in network security.

The security mechanisms needed to cope with unwanted access fall into two broad categories (see [Figure 1.6](#)). The first category might

be termed a gatekeeper function. It includes password-based login procedures that are designed to deny access to all but authorized users and screening logic that is designed to detect and reject worms, viruses, and other similar attacks. Once either an unwanted user or

unwanted software gains access, the second line of defense consists of a variety of internal controls that monitor activity and analyze stored information in an attempt to detect the presence of unwanted intruders. These issues are explored in [Part Four](#).

Plaintext: This is the original intelligible message or data that is fed into the algorithm as input.

Encryption algorithm: The encryption algorithm performs various substitutions and transformations on the plaintext.

Secret key: The secret key is also input to the encryption algorithm. The key is a value independent of the plaintext and of the algorithm. The algorithm will produce a different output depending on the specific key being used at the time. The exact substitutions and transformations performed by the algorithm depend on the key.

Information Security and Cryptography

Ciphertext: This is the scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts. The ciphertext is an apparently random stream of data and, as it stands, is unintelligible.

Decryption algorithm: This is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.

Stream Ciphers and Block Ciphers

A **stream cipher** is one that encrypts a digital data stream one bit or one byte at a time. Examples of classical stream ciphers are the autokeyed Vigenère cipher and the Vernam cipher.

A **block cipher** is one in which a block of plaintext is treated as a whole and used to produce a ciphertext block of equal length. Typically, a block size of 64 or 128 bits is used. Using some of the modes of operation explained in [Chapter 6](#), a block cipher can be used to achieve the same effect as a stream cipher. Far more effort has gone into analyzing block ciphers. In general, they seem applicable to a broader range of applications than stream ciphers. The vast majority of network-based symmetric cryptographic applications make use of block ciphers. Accordingly, the concern in this chapter, and in our discussions throughout the book of symmetric encryption, will focus on block ciphers.

Note:

Plaintext → small letter	Key → capital letter	Cipher text → capital letter
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Decipher

Cipher text → capital letter	Key → small letter	Plain text → small letter
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Block: A sequence of consecutive characters encoded at one time

block length: The number of characters in a block

An algorithm for performing encryption (and the reverse, decryption) - a series of well-defined steps that can be followed as a procedure. Works at the level of individual letters, or small groups of letters

Ciphertext: A text in the encrypted form produced by some cryptosystem. The convention is for ciphertexts to contain no white space or punctuation

Cryptanalysis: The analysis and deciphering of cryptographic writings or systems.

cryptography: The process or skill of communicating in or deciphering secret writings or ciphers

cryptosystem: The package of all processes, formulae, and instructions for encoding and decoding messages using cryptography

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decryption: Any procedure used in cryptography to convert cipher- text (encrypted data) into plaintext

digram: Sequence of two consecutive characters

encryption: The process of putting text into encoded form

key: A relatively small amount of information that is used by an algorithm to customize the transformation of plaintext into ciphertext (during encryption) or vice versa (during decryption)

key length: The size of the key - how many values comprise the key

monoalphabetic: Using one alphabet - refers to a cryptosystem where each alphabetic character is mapped to a unique alphabetic character

one-time pad: Another name for the Vernam cipher

plaintext: A message before encryption or after decryption, i.e., in its usual form which anyone can read, as opposed to its encrypted form

polyalphabetic: Using many alphabets - refers to a cipher where each alphabetic character can be mapped to one of many possible alphabetic characters

trigram: Sequence of three consecutive characters unigram

The field of Cryptology today represents that branch of information theory which deals with the security of information confidentiality. Methods in cryptology may be subdivided into two classes, namely that of **cryptography** (methods applied by authorized information sharers to design and develop encryption schemes in order to ensure confidentiality of information) and that of **cryptanalysis** (mathematical and statistical attempts by unauthorized persons to break cipher in order to reveal the meaning of the underlying protected data)[1].

Cryptography may in fact be sub classified into **Symmetric** and **Asymmetric** cryptography ciphers. Symmetric cryptography ciphers sub classified into block ciphers and stream cipher.

In **Symmetric** cryptography ciphers the enciphering and deciphering keys are the same, as shown in Fig. 1.1.

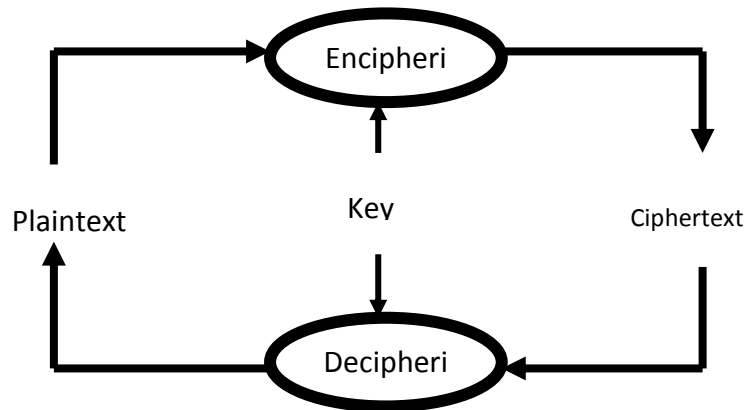


Fig. 1.1: Secret Writing

Block ciphers, in which blocks of data, known as plaintext, are transformed into cipher text which appears unintelligible to unauthorized persons.

Stream ciphers, which involve streams of typically binary operations and are well suited for efficient computer implementation.

The **ciphertext** is created by choosing a permutation of the 26-character alphabet and using it to replace each letter in the *plaintext* message.

Lec. 2

Type of Ciphers:

Encryption and Authentication

Nearly all modern security mechanisms are based on keeping secrets private to certain individuals. Security systems use encryption to keep secrets, and they use authentication to prove the identity of individuals. These two basic security mechanisms are the foundation upon which nearly all security mechanisms are based.

- 1-Secret key encryption
- 2-Hashes and one-way functions
- 3-Public key encryption
- 4-Password authentication
- 5-Challenge/response authentication
- 6-Sessions
- 7-Public key authentication
- 8-Digital signatures
- 9-Certificates
- 10-Biometric authentication

Data security is the sciences and study of methods of protecting data in computer and communication systems for unauthorized disclosure and modification.

Or

* **Cipher** is a secret method of writing.

1- Caser cipher

It is believed that Julius Caesar, in the period 58 BCE to 51 BCE, enciphered messages to his lawyer Marcus Tullius Cicero and other Roman senators using a monoalphabetic substitution.

In the Caesar cipher, each plaintext letter was replaced by the letter standing three places to-the-right in the alphabet. If we neglect that the original Roman or Latin alphabet did not contain a J, U, or W, then, Julius' query in the present day Roman alphabet.

A	B	C	D	E	F	G	H	I	J	K
0	1	2	3	4	5	6	7	8	9	10
L	M	N	O	P	Q	R	S	T	U	V
11	12	13	14	15	16	17	18	19	20	21
W	X	Y	Z							
22	23	24	25							

Example: (Caser method) use $k=3$

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Plaintext: Ali

(Encipher process)..... **(P+K) mod 26**

Sol:

A=0

$0+3 \bmod 26 = 3 = D$

L=11

$11+3 \bmod 26 = 14 = O$

I=8

$8+3 \bmod 26 = 11 = L$

Cipher text: DOL

(Decipher process)..... **(C-K) mod 26**

Sol:

D=3

$3-3 \bmod 26 = 0 = A$

O=14

$14-3 \bmod 26 = 11 = L$

L=11

$11-3 \bmod 26 = 8 = I$

EX: 2

Plain text; anyone know where i can get decent pizza?

Cipher text: DQBRQH NQRZ ZKHUH L EDQ JHW GHFHQW SLCCD?

Plaintext letter	A	B	C	D		W	X	Y	Z
Ciphertext letter	D	E	F	G	...	Z	A	B	C

The earliest known use of a substitution cipher, and the simplest, was by Julius Caesar. The Caesar cipher involves replacing each letter of the alphabet with the letter standing three places further down the alphabet. For example,

Plain : meet me after the toga party

Cipher: PHHW PH DIWHU WKH WRJD SDUWB

Note that the alphabet is wrapped around, so that the letter following Z is A. We can define the transformation by listing all possibilities, as follows:

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z

cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Let us assign a numerical equivalent to each letter:

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Then the algorithm can be expressed as follows. For each plaintext letter p , substitute the ciphertext letter C :

We define $a \bmod n$ to be the remainder when a is divided by n . For example, $11 \bmod 7 = 4$.

$$C = E(3, p) = (p + 3) \bmod 26$$

A shift may be of any amount, so that the general Caesar algorithm is

$$C = E(k, p) = (p + k) \bmod 26$$

where k takes on a value in the range 1 to 25. The decryption algorithm is simply

$$p = D(k, C) = (C - k) \bmod 26$$

2- Substitution Ciphers

Substitution ciphers are block ciphers which replace symbols (or groups of symbols) by other symbols or groups of symbols.

Mono alphabetic substitution ciphers

The simplest cipher of those attacked is the mono alphabetic substitution cipher.

This cipher is described as follows:

- Let the plaintext and the cipher text character sets be the same, say the alphabet Z
- Let the keys, K , consist of all possible permutations of the symbols in Z
- For each permutation $\pi \in K$,
 1. define the encryption function $e_{\pi}(x) = \pi(x)$
 2. and define the decryption function $d_{\pi}(y) = \pi^{-1}(y)$, where π^{-1} is the inverse permutation to π .

For example, define Z to be the 26 letter English alphabet. Then, a random permutation π could be (plaintext characters are in lower case and cipher text characters in upper case) as shown in figure 1 .

Plaintext:
upon this basis i am going to show you how a bunch of bright young folks did find a champion a man with boys and girls of his own.
Key:
G A B S L Y T E X U C F H I J K Z M N O P Q R D V W

Figure 1: Example of a key of a single character substitution cipher.

Substitution Ciphers

1.1 Mono alphabetic

1.2 Poly alphabetic

Substitution Cipher:-

Is a symmetric cryptography ciphers, a substitution cipher is classify into two part (Mono alphabetic & Poly alphabetic).

In a substitution ciphers the value of character or character string is changed when transforming the plaintext into cipher text, but the position of the original string and its value replacement correspond exactly in the plain and cipher texts.

Plain Text Character:-

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Cipher Text Character (key cipher)

H	W	U	G	C	T	V	A	E	K	D	Y	Q	P	B	R	J	L	F	I	X	M	S	O	Z	N
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Let the Plain text = "dulce et decorm est pro patria mori"

Now the cipher text is GXYUCCIGCUBLXQCFIRLBRHILEHQBLE

Note:- Remove all spaces, special characters in plain text

Example:

Keyword= CRYPTOGRAPHIC SYSTEM

P: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

K: C R Y P T O G A H I S E M B D F J K L N Q U V W X Z

To encipher the plaintext= cryptography

ciphertext=YKXFNDGKCFAX

3-Polyalphabetic Ciphers

Another way to improve on the simple monoalphabetic technique is to use different monoalphabetic substitutions as one proceeds through the plaintext message. The general name for this approach is **polyalphabetic substitution cipher**. All these techniques have the

following features in common:

1. A set of related monoalphabetic substitution rules is used.
2. A key determines which particular rule is chosen for a given transformation.

The best known, and one of the simplest, such algorithm is referred to as the Vigenère cipher. In this scheme, the set of related monoalphabetic substitution rules consists of the 26 Caesar ciphers, with shifts of 0 through 25. Each cipher is denoted by a key letter, which is the ciphertext letter that substitutes for the plaintext letter a. Thus, a Caesar cipher with a shift of 3 is denoted by the key value d .

To aid in understanding the scheme and to aid in its use, a matrix known as the Vigenère tableau is constructed (Table 2.3). Each of the 26 ciphers is laid out horizontally, with the key letter for each cipher to its left. A normal alphabet for the plaintext runs across the top. The process of encryption is simple: Given a key letter x and a plaintext letter y , the ciphertext letter is at the intersection of the row labeled x and the column labeled y ; in this case the ciphertext is V.

3-1Poly alphabetic substitution ciphers

The poly alphabetic substitution cipher is a simple extension of the mono alphabetic one. The difference is that the message is broken into blocks of equal length, say B , and then each position in the block ($1 \dots B$) is encrypted (or decrypted) using a different simple substitution cipher key. The block size (B) is often referred to as the period of the cipher.

All these techniques have the following features in common.

1. A set of related mono alphabetic substitution rules is used.
2. A key determines which particular rule is chosen for a given transformation[13, 21].

The best known, and one of the simplest, such algorithm is referred to as the Vigenère cipher. In this scheme, the set of related mono alphabetic substitution rules consists of the 26 Caesar ciphers, with shifts of 0 through 25 as shown in Table 1. Each cipher is denoted by a key letter, which is the cipher text letter that substitutes for the plaintext letter a. Thus, a Caesar cipher with a shift of 3 is denoted by the key value d . To aid in understanding the scheme and to aid in its use, a matrix known as the Vigenère tableau is constructed (Table.3. Each of the 26 ciphers is laid out horizontally, with the key letter for each cipher to its left. A normal alphabet for the plaintext runs across the top. The process of encryption is simple: Given a key letter x and a plaintext letter y , the cipher text letter is at the intersection of the row labeled x and the column labeled y ; in this case the cipher text is V. Table 2 shows the cipher text in Vigenere method.

Table (1): Mapping Letters To Integers And Back

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

The encryption function seen in Equation 2.

$$E(k, m_i) = m_i + k \pmod{26} \dots\dots(2)$$

Table (2): Vigenere Cipher Encryption Example

Keyword:																													
s	U	B	S	T	I	T	U	T	I	O	N	S	U	B	S	T	I	T	U	T	I	O	N	S	U	B	S	T	I
As integer																													
18	20	1	18	19	8	19	20	19	8	14	13	18	20	1	18	19	8	19	20	19	8	14	13	18	19	1	18	19	8
Plaintext:																													
t	o	b	e	o	r	n	o	t	t	o	b	e	t	h	a	t	i	s	t	h	e	q	u	e	s	t	i	o	n
As integer																													
19	14	1	4	14	17	13	14	19	19	14	1	4	19	7	0	19	8	18	19	7	4	16	20	4	18	19	8	14	13
Addition																													
11	8	2	22	7	25	6	8	12	1	2	14	23	13	8	18	12	16	11	13	0	12	4	7	23	12	20	0	7	21
Cipher text:																													
L	I	C	W	H	Z	G	I	M	B	C	O	W	N	I	S	M	Q	L	N	A	M	E	H	W	M	U	A	H	V

The Vigenere Tableau

The Vigenere Cipher, proposed by Blaise de Vigenere from the court of Henry III of France in the sixteenth century, is a poly alphabetic substitution based on the following tableau(Table 3).

For example, suppose we wish to encipher the plaintext message: "TO BE OR NOT TO BE THAT IS THE QUESTION", using the keyword SUBSTITUTION. We begin by writing the keyword, repeated as many times as necessary, above the plaintext message. To derive the ciphertext using the tableau, for each letter in the plaintext, one finds the intersection of the row given by the corresponding keyword letter and the column given by the plaintext letter itself to pick out the cipher text letter[13, 23].

Table 3: The Vigenere Tableau

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	G
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Lec. 3

4-Hill Cipher

Introduction

The computation used in the Hill cipher is based on linear algebra techniques. As time progressed, the study of cryptography continued to mature and, more recently, began to involve higher level mathematics. With this more advanced math came more advanced ciphers based on the idea of encryption and decryption keys. Encryption keys are a special value or set of values used in an encryption algorithm to convert a plaintext into a cipher text. A decryption key is the opposite. Decryption keys are used as part of a decryption algorithm to convert the cipher text back into the original plaintext.

Table (1): inverse of a

1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

Encryption Low

$$C = (k * p) \bmod 26$$

Decryption Low

$$P = (k_{-1} * C) \bmod 26$$

- C is the cipher text
- k is the key matrix or encryption matrix
- k₋₁ inverse key matrix or decryption matrix
- P is the plaintext

Example 1

Use Hill cipher method to encrypt the plaintext *"TOP SECRET MESSAGE"* with

encryption matrix $\begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix}$

Answer

•Encryption process with the Hill Cipher:

Encrypting text using the Hill cipher is accomplished by breaking a given plaintext into blocks of size n (where n is an integer), writing these blocks as column vectors, if you have an odd number of

letters, repeat the last letter, and then multiplying these column vectors by any invertible $n \times n$ matrix.

The encryption matrix must be invertible because its inverse will be used to decrypt the cipher texts created with the Hill cipher and this encryption matrix. The invertibility of the encryption matrix allows us to say that its determinant must not be 0. The determinant of the encryption matrix must also be relatively prime to the size of the alphabet.

Compute the determinate of the matrix

This matrix has the determinant $(3*7) - (2*5) = 21 - 10 = 11$. Since 11 is $\neq 0$, this matrix is invertible. 11 is also relatively prime to 26. These two qualities satisfy the requirements listed previously, making this encryption matrix a valid choice for use in the Hill cipher

Split the plaintext into blocks of size 2 (ignoring spaces), determine the letters' numerical values, and align these as column vectors. If the length of the plaintext is not evenly divisible by 2, add a previously decided character to the end of the string until the plaintext is evenly divisible by 2.

Divide the message (top secret message) to pairs

$$\begin{array}{l} \begin{bmatrix} T \\ O \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} \quad \begin{bmatrix} P \\ S \end{bmatrix} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} \quad \begin{bmatrix} E \\ C \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \quad \begin{bmatrix} R \\ E \end{bmatrix} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} \quad \begin{bmatrix} T \\ M \end{bmatrix} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} \\ \begin{bmatrix} E \\ S \end{bmatrix} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} \quad \begin{bmatrix} S \\ A \end{bmatrix} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} \quad \begin{bmatrix} G \\ E \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} \end{array}$$

Multiply each of these column vectors by the encryption matrix and take mod 26 of the result. The encryption formula for hill method is

$$C = (k * p) \bmod 26$$

top secret message

$$C = (k * p) \bmod 26$$

$$\begin{bmatrix} T \\ O \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 19 \\ 14 \end{vmatrix} = \begin{vmatrix} (3 * 19) + (2 * 14) \\ (5 * 19) + (7 * 14) \end{vmatrix} = \begin{vmatrix} 85 \\ 193 \end{vmatrix} \pmod{26} = \begin{vmatrix} 7 \\ 11 \end{vmatrix} = \begin{vmatrix} H \\ L \end{vmatrix}$$

top secret message

$$C = (k * p) \pmod{26}$$

$$\begin{vmatrix} P \\ S \end{vmatrix} = \begin{vmatrix} 15 \\ 18 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 15 \\ 18 \end{vmatrix} = \begin{vmatrix} (3 * 15) + (2 * 18) \\ (5 * 15) + (7 * 18) \end{vmatrix} = \begin{vmatrix} 81 \\ 201 \end{vmatrix} \pmod{26} = \begin{vmatrix} 3 \\ 19 \end{vmatrix} = \begin{vmatrix} D \\ T \end{vmatrix}$$

top secret message

$$C = (k * p) \pmod{26}$$

$$\begin{vmatrix} E \\ C \end{vmatrix} = \begin{vmatrix} 4 \\ 2 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 4 \\ 2 \end{vmatrix} = \begin{vmatrix} (3 * 4) + (2 * 2) \\ (5 * 4) + (7 * 2) \end{vmatrix} = \begin{vmatrix} 16 \\ 34 \end{vmatrix} \pmod{26} = \begin{vmatrix} 16 \\ 8 \end{vmatrix} = \begin{vmatrix} Q \\ I \end{vmatrix}$$

top secret message

$$C = (k * p) \pmod{26}$$

$$\begin{vmatrix} R \\ E \end{vmatrix} = \begin{vmatrix} 17 \\ 4 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 17 \\ 4 \end{vmatrix} = \begin{vmatrix} (3 * 17) + (2 * 4) \\ (5 * 17) + (7 * 4) \end{vmatrix} = \begin{vmatrix} 59 \\ 113 \end{vmatrix} \pmod{26} = \begin{vmatrix} 7 \\ 9 \end{vmatrix} = \begin{vmatrix} H \\ J \end{vmatrix}$$

top secret message

$$\begin{vmatrix} T \\ M \end{vmatrix} = \begin{vmatrix} 19 \\ 12 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 19 \\ 12 \end{vmatrix} = \begin{vmatrix} (3 * 19) + (2 * 12) \\ (5 * 19) + (7 * 12) \end{vmatrix} = \begin{vmatrix} 81 \\ 179 \end{vmatrix} \pmod{26} = \begin{vmatrix} 3 \\ 23 \end{vmatrix} = \begin{vmatrix} D \\ X \end{vmatrix}$$

$$\begin{vmatrix} E \\ S \end{vmatrix} = \begin{vmatrix} 4 \\ 18 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 4 \\ 18 \end{vmatrix} = \begin{vmatrix} (3 * 4) + (2 * 18) \\ (5 * 4) + (7 * 18) \end{vmatrix} = \begin{vmatrix} 48 \\ 146 \end{vmatrix} \pmod{26} = \begin{vmatrix} 22 \\ 16 \end{vmatrix} = \begin{vmatrix} W \\ Q \end{vmatrix}$$

$$\begin{vmatrix} S \\ A \end{vmatrix} = \begin{vmatrix} 18 \\ 0 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 18 \\ 0 \end{vmatrix} = \begin{vmatrix} (3 * 18) + (2 * 0) \\ (5 * 18) + (7 * 0) \end{vmatrix} = \begin{vmatrix} 54 \\ 90 \end{vmatrix} \pmod{26} = \begin{vmatrix} 2 \\ 12 \end{vmatrix} = \begin{vmatrix} C \\ M \end{vmatrix}$$

$$\begin{vmatrix} G \\ E \end{vmatrix} = \begin{vmatrix} 6 \\ 4 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 6 \\ 4 \end{vmatrix} = \begin{vmatrix} (3 * 6) + (2 * 4) \\ (5 * 6) + (7 * 4) \end{vmatrix} = \begin{vmatrix} 26 \\ 58 \end{vmatrix} \pmod{26} = \begin{vmatrix} 0 \\ 6 \end{vmatrix} = \begin{vmatrix} A \\ G \end{vmatrix}$$

Cipher text:

“HLD TQIHJDXWQCMAG”

Decryption process

•We are interested in how the party receiving a secret message encoded by the Hill cipher can decode it into the original plaintext. As previously described, the Hill cipher is based on matrix multiplication and any encryption matrix used in the Hill cipher must be invertible. The process is the same as encryption, but with the inverse matrix instead of the original encryption matrix.

•Decryption of the cipher text “HLDTQIHJDXWQCMAG” with the 2x2 encryption matrix previously defined would go as follows:

Example 2

Use hill cipher method to decryption the cipher text

“HLDTQIHJDXWQCMAG” with the 2x2 key

**encryption [3 2
5 7]**

Split the cipher text into blocks of 2, determine the letters' numerical values, and align these as column vectors.

$$\begin{array}{ccccc} \begin{bmatrix} H \\ L \end{bmatrix} = \begin{bmatrix} 7 \\ 11 \end{bmatrix} & \begin{bmatrix} D \\ T \end{bmatrix} = \begin{bmatrix} 3 \\ 19 \end{bmatrix} & \begin{bmatrix} Q \\ I \end{bmatrix} = \begin{bmatrix} 16 \\ 8 \end{bmatrix} & \begin{bmatrix} H \\ J \end{bmatrix} = \begin{bmatrix} 7 \\ 9 \end{bmatrix} & \begin{bmatrix} D \\ X \end{bmatrix} = \begin{bmatrix} 3 \\ 23 \end{bmatrix} \\ \begin{bmatrix} W \\ Q \end{bmatrix} = \begin{bmatrix} 22 \\ 16 \end{bmatrix} & \begin{bmatrix} C \\ M \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \end{bmatrix} & \begin{bmatrix} A \\ G \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix} & & \end{array}$$

Find the inverse key

$$\det\left(\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix}\right) = (3*7) - (2*5) = 11$$

$$11^{-1} \bmod 26 = 19$$

$$19 \begin{vmatrix} 7 & -2 \\ -5 & 3 \end{vmatrix} = \begin{vmatrix} 133 & -38 \\ -95 & 57 \end{vmatrix} \pmod{26} = \begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix}$$

$$\mathbf{K}^{-1} = \begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix}$$

Multiply each of these column vectors above by the decryption matrix calculated in step 1 and take mod 26 of the result. The decryption formula for hill method is

$$P = (k^{-1} * C) \bmod 26$$

HLDTQIHJDXWQCMAG|

$$P = (k^{-1} * C) \bmod 26$$

$$\begin{vmatrix} H \\ L \end{vmatrix} = \begin{vmatrix} 7 \\ 11 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 7 \\ 11 \end{vmatrix} = \begin{vmatrix} (3 * 7) + (14 * 11) \\ (9 * 7) + (5 * 11) \end{vmatrix} = \begin{vmatrix} 175 \\ 118 \end{vmatrix} \bmod 26 = \begin{vmatrix} 19 \\ 14 \end{vmatrix} = \begin{vmatrix} T \\ O \end{vmatrix}$$

$$\begin{vmatrix} D \\ T \end{vmatrix} = \begin{vmatrix} 3 \\ 19 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 3 \\ 19 \end{vmatrix} = \begin{vmatrix} (3 * 3) + (14 * 19) \\ (9 * 3) + (5 * 19) \end{vmatrix} = \begin{vmatrix} 275 \\ 122 \end{vmatrix} \bmod 26 = \begin{vmatrix} 15 \\ 18 \end{vmatrix} = \begin{vmatrix} P \\ S \end{vmatrix}$$

$$\begin{vmatrix} Q \\ I \end{vmatrix} = \begin{vmatrix} 16 \\ 8 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 16 \\ 8 \end{vmatrix} = \begin{vmatrix} (3 * 16) + (14 * 8) \\ (9 * 16) + (5 * 8) \end{vmatrix} = \begin{vmatrix} 160 \\ 184 \end{vmatrix} \bmod 26 = \begin{vmatrix} 4 \\ 2 \end{vmatrix} = \begin{vmatrix} E \\ C \end{vmatrix}$$

$$\begin{vmatrix} H \\ J \end{vmatrix} = \begin{vmatrix} 7 \\ 9 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 7 \\ 9 \end{vmatrix} = \begin{vmatrix} (3 * 7) + (14 * 9) \\ (9 * 7) + (5 * 9) \end{vmatrix} = \begin{vmatrix} 147 \\ 108 \end{vmatrix} \bmod 26 = \begin{vmatrix} 17 \\ 4 \end{vmatrix} = \begin{vmatrix} R \\ E \end{vmatrix}$$

$$\begin{vmatrix} D \\ X \end{vmatrix} = \begin{vmatrix} 3 \\ 23 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 3 \\ 23 \end{vmatrix} = \begin{vmatrix} (3 * 3) + (14 * 23) \\ (9 * 3) + (5 * 23) \end{vmatrix} = \begin{vmatrix} 331 \\ 142 \end{vmatrix} \bmod 26 = \begin{vmatrix} 19 \\ 12 \end{vmatrix} = \begin{vmatrix} T \\ M \end{vmatrix}$$

$$\begin{vmatrix} W \\ Q \end{vmatrix} = \begin{vmatrix} 22 \\ 16 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 22 \\ 16 \end{vmatrix} = \begin{vmatrix} (3 * 22) + (14 * 16) \\ (9 * 22) + (5 * 16) \end{vmatrix} = \begin{vmatrix} 290 \\ 278 \end{vmatrix} \bmod 26 = \begin{vmatrix} 4 \\ 18 \end{vmatrix} = \begin{vmatrix} E \\ S \end{vmatrix}$$

$$\begin{vmatrix} C \\ M \end{vmatrix} = \begin{vmatrix} 2 \\ 12 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 2 \\ 12 \end{vmatrix} = \begin{vmatrix} (3 * 2) + (14 * 12) \\ (9 * 2) + (5 * 12) \end{vmatrix} = \begin{vmatrix} 174 \\ 78 \end{vmatrix} \bmod 26 = \begin{vmatrix} 18 \\ 0 \end{vmatrix} = \begin{vmatrix} S \\ A \end{vmatrix}$$

$$\begin{vmatrix} A \\ G \end{vmatrix} = \begin{vmatrix} 0 \\ 6 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 14 \\ 9 & 5 \end{vmatrix} \begin{vmatrix} 0 \\ 6 \end{vmatrix} = \begin{vmatrix} (3 * 0) + (14 * 6) \\ (9 * 0) + (5 * 6) \end{vmatrix} = \begin{vmatrix} 84 \\ 30 \end{vmatrix} \bmod 26 = \begin{vmatrix} 6 \\ 4 \end{vmatrix} = \begin{vmatrix} G \\ E \end{vmatrix}$$

Decryption process

We are take the character above we get the
Original plaintext:

"TOP SECRET MESSAGE"

Example 3 / In a Hill cipher decrypt the ciphertext
“VOHY” with key $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

Answer

Step1 Find the key inverse

Determinant : $\begin{bmatrix} d & b \\ c & a \end{bmatrix}$ determinant is calculated by $ad - bc$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} = (3*5) - (3*2) = 9 \implies \text{inverse equal } 3$$

$$3 * \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 15 & -9 \\ -6 & 9 \end{bmatrix} \mod 26 = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

Example 3 / In a Hill cipher decrypt the ciphertext

“VOHY” with key $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

Answer

Step2 The encryption process

$$\begin{bmatrix} V \\ O \end{bmatrix} = \begin{bmatrix} 21 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} H \\ Y \end{bmatrix} = \begin{bmatrix} 7 \\ 24 \end{bmatrix}$$

$$P = (k^{-1} * C) \mod 26$$

$$\begin{bmatrix} V \\ O \end{bmatrix} \implies \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 21 \\ 14 \end{bmatrix} = \begin{bmatrix} 15 * 21 + 17 * 14 \\ 20 * 21 + 9 * 14 \end{bmatrix} = \begin{bmatrix} 553 \\ 546 \end{bmatrix} \mod 26 = \begin{bmatrix} 7 \\ 0 \end{bmatrix} = HA$$

$$\begin{bmatrix} H \\ Y \end{bmatrix} \implies \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 24 \end{bmatrix} = \begin{bmatrix} 15 * 7 + 17 * 24 \\ 20 * 7 + 9 * 24 \end{bmatrix} = \begin{bmatrix} 513 \\ 356 \end{bmatrix} \mod 26 = \begin{bmatrix} 19 \\ 18 \end{bmatrix} = TS$$

Plaintext = HATS

Example 4 / If the key is $\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$ and the ciphertext is “RXAODY” find the plaintext using Hill cipher method.

Answer

Step1 Find the key inverse

find the determent $d = (24 * 14) - (5 * 19) = 241$

$$d = 241 \bmod 26 = 7$$

$$d^{-1} = 15$$

$$k^{-1} = d^{-1}(k) * = 15 * \begin{pmatrix} 14 & -19 \\ -5 & 24 \end{pmatrix} = \begin{pmatrix} 210 & -285 \\ -75 & 360 \end{pmatrix} \bmod 26$$

$$k^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 22 \end{pmatrix}$$

Example 4 / If the key is $\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$ and the ciphertext is “RXAODY” find the plaintext using Hill cipher method.

Answer

Step2 The encryption process

$$P = (k^{-1} * C) \bmod 26$$

$$\bullet \text{ RX} = \begin{pmatrix} 17 \\ 23 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 3 & 22 \end{pmatrix} \begin{pmatrix} 17 \\ 23 \end{pmatrix} = \begin{pmatrix} 57 \\ 557 \end{pmatrix} \bmod 26 = \begin{pmatrix} 5 \\ 11 \end{pmatrix} \rightarrow \text{FL}$$

Example 4 / If the key is $\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$ and the ciphertext is “RXAODY” find the plaintext using Hill cipher method.

Answer

Step2 The encryption process

- $AO = \begin{pmatrix} 0 \\ 14 \end{pmatrix}$

$$\begin{pmatrix} 2 & 1 \\ 3 & 22 \end{pmatrix} \begin{pmatrix} 0 \\ 14 \end{pmatrix} = \begin{pmatrix} 14 \\ 308 \end{pmatrix} \bmod 26 = \begin{pmatrix} 14 \\ 22 \end{pmatrix} \rightarrow OW$$

Example 4 / If the key is $\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$ and the ciphertext is “RXAODY” find the plaintext using Hill cipher method.

Answer

Step2 The encryption process

- $DY = \begin{pmatrix} 3 \\ 24 \end{pmatrix}$

$$\begin{pmatrix} 2 & 1 \\ 3 & 22 \end{pmatrix} \begin{pmatrix} 3 \\ 24 \end{pmatrix} = \begin{pmatrix} 30 \\ 537 \end{pmatrix} \bmod 26 = \begin{pmatrix} 4 \\ 17 \end{pmatrix} \rightarrow ER$$

plaintext : **flower.**

■ **Message:** ATTACK IS TONIGHT

■ **Key** =
$$\begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix}$$

The **first step** is to convert the given keyword to a 3x3 matrix form. Next, convert the keyword matrix into a key matrix by replacing the letters with corresponding numeric values.

■ **Message:** ATTACK IS TONIGHT

Assign : A-Z 0-25

$$\begin{bmatrix} A & T & T \\ A & C & K \\ I & S & T \\ O & N & I \\ G & H & T \end{bmatrix} = \begin{bmatrix} 0 & 19 & 19 \\ 0 & 2 & 10 \\ 8 & 18 & 19 \\ 14 & 13 & 8 \\ 6 & 7 & 19 \end{bmatrix}$$

■ **Message:** ATTACK IS TONIGHT

■ **Cipher Text = (Plain Text x Key) Mod 26**

$$= \begin{bmatrix} 0 & 19 & 19 \\ 0 & 2 & 10 \\ 8 & 18 & 19 \\ 14 & 13 & 8 \\ 6 & 7 & 19 \end{bmatrix} \times \begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix} \text{ Mod } 26$$

$$\begin{bmatrix} 0 & 19 & 19 \\ 0 & 2 & 10 \\ 8 & 18 & 19 \\ 14 & 13 & 8 \\ 6 & 7 & 19 \end{bmatrix} \times \begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix} \text{ Mod } 26$$

$$C_{11} = 0*3 + 19*20 + 19*9 = 551 \text{ Mod } 26 = 05$$

$$C_{12} = 0*10 + 19*9 + 19*4 = 247 \text{ Mod } 26 = 13$$

$$C_{13} = 0*20 + 19*17 + 19*17 = 646 \text{ Mod } 26 = 22$$

ATT → **FNW** Similarly you can calculate other...

Decryption

$$\begin{aligned} \bullet \text{ Det (Key)} &= \begin{vmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{vmatrix} \text{ Mod } 26 = 03 \\ &= 3*(9*17 - 17*4) - 10*(20*17 - 17*9) + 20*(20*4 - 9*9) \\ &= (-1635) \text{ Mod } 26 = (-23) \text{ Mod } 26 = 03 \\ [Det (Key)]^{-1} &= 03^{-1} \text{ Mod } 26 = 09 \end{aligned}$$

$$\text{Key} = \begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix}$$

$$\text{Trans(Key)} = \begin{bmatrix} 3 & 20 & 9 \\ 10 & 9 & 4 \\ 20 & 17 & 17 \end{bmatrix}$$

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$$\bullet \text{ Trans(Key)} = \begin{bmatrix} 3 & 20 & 9 \\ 10 & 9 & 4 \\ 20 & 17 & 17 \end{bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \text{ To find Minor of}$$

$$a_{11} = a_{22} \cdot a_{33} - a_{32} \cdot a_{23} = 85$$

$$\begin{aligned} a_{11} &= 85 & a_{12} &= 90 & a_{13} &= (-10) \\ a_{21} &= 187 & a_{22} &= (-129) & a_{23} &= (-349) \\ a_{31} &= (-1) & a_{32} &= (-78) & a_{33} &= (-173) \end{aligned}$$

$$\text{Minor} = \begin{bmatrix} 85 & 90 & -10 \\ 187 & -129 & -349 \\ -1 & -78 & -173 \end{bmatrix}$$

$$\text{Minor} = \begin{bmatrix} 85 & 90 & -10 \\ 187 & -129 & -349 \\ -1 & -78 & -173 \end{bmatrix}$$

Put Sign According to $(-1)^{i+j}$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \rightarrow \begin{bmatrix} 85 & -90 & -10 \\ -187 & -129 & 349 \\ -1 & 78 & -173 \end{bmatrix}$$

$$\text{Key}^{-1} = [\text{Det (Key)}]^{-1} \times \text{Adj (Key)}$$

$$= 09 * \begin{bmatrix} 85 & -90 & -10 \\ -187 & -129 & 349 \\ -1 & 78 & -173 \end{bmatrix} \text{Mod } 26$$

$$= \begin{bmatrix} 765 & -810 & -90 \\ -1683 & -1161 & 3141 \\ -9 & 702 & -1557 \end{bmatrix} \text{Mod } 26 = \begin{bmatrix} 11 & 22 & 14 \\ 7 & 9 & 21 \\ 17 & 0 & 3 \end{bmatrix}$$

Table (1): inverse of a											
1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

<https://www.cliffsnotes.com/study-notes/7281430>

This cipher is somewhat more difficult to understand than the others in this chapter, but it illustrates an important point about cryptanalysis that will be useful later on. This subsection can be skipped on a first reading.

Another interesting multi letter cipher is the Hill cipher, developed by the mathematician Lester Hill in 1929. The encryption algorithm takes m successive plaintext letters and substitutes for them m ciphertext letters. The substitution is determined by m linear equations in which each character is assigned a numerical value ($a = 0, b = 1 \dots z = 25$). For $m = 3$, the system can be described as follows:

$$\begin{aligned} c_1 &= (k_{11}P_1 + k_{12}P_2 + k_{13}P_3) \bmod 26 \\ c_2 &= (k_{21}P_1 + k_{22}P_2 + k_{23}P_3) \bmod 26 \\ c_3 &= (k_{31}P_1 + k_{32}P_2 + k_{33}P_3) \bmod 26 \end{aligned}$$

This can be expressed in term of column vectors and matrices:

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} \bmod 26$$

$$\mathbf{C} = \mathbf{K}\mathbf{P} \bmod 26$$

where $\mathbf{C}$ and $\mathbf{P}$ are column vectors of length 3, representing the plaintext and ciphertext, and $\mathbf{K}$ is a 3 x 3 matrix, representing the encryption key. Operations are performed mod 26.

For example, consider the plaintext "paymoremoney" and use the encryption key

$$\mathbf{K} = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

The first three letters of the plaintext are represented by the vector

$$\begin{pmatrix} 15 \\ 0 \\ 24 \end{pmatrix}. \text{ Then } \mathbf{K} \begin{pmatrix} 15 \\ 0 \\ 24 \end{pmatrix} = \begin{pmatrix} 375 \\ 819 \\ 486 \end{pmatrix} \bmod 26 = \begin{pmatrix} 11 \\ 13 \\ 18 \end{pmatrix} = \text{LNS. Continuing in this fashion,}$$

the ciphertext for the entire plaintext is LNSHDLEWMTRW.

Decryption requires using the inverse of the matrix $\mathbf{K}$. The inverse $\mathbf{K}^{-1}$ of a matrix $\mathbf{K}$ is defined by the equation $\mathbf{K}\mathbf{K}^{-1} = \mathbf{K}^{-1}\mathbf{K} = \mathbf{I}$, where $\mathbf{I}$ is the matrix that is all zeros except for ones along the main diagonal from upper left to lower right. The inverse of a matrix does not always exist, but when it does, it satisfies the preceding equation. In this case, the inverse is:

$$\mathbf{K}^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix}$$

This is demonstrated as follows:

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} = \begin{pmatrix} 443 & 442 & 442 \\ 858 & 495 & 780 \\ 494 & 52 & 365 \end{pmatrix} \bmod 26 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

It is easily seen that if the matrix $\mathbf{K}^{-1}$ is applied to the ciphertext, then the plaintext is recovered. To explain how the inverse of a matrix is determined, we make an exceedingly brief excursion into linear algebra.

For any square matrix ($m \times m$) the **determinant** equals the sum of all the products that can be formed by taking exactly one element from each row and exactly one element from each column, with certain of the product terms preceded by a minus sign. For a 2×2 matrix

$$\begin{pmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{pmatrix}$$

the determinant is $k_{11}k_{22} - k_{12}k_{21}$. For a 3×3 matrix, the value of the determinant is $k_{11}k_{22}k_{33} + k_{21}k_{32}k_{13} + k_{31}k_{12}k_{23} - k_{31}k_{22}k_{13} - k_{21}k_{12}k_{33} - k_{11}k_{32}k_{23}$.

If a square matrix $\mathbf{A}$ has a nonzero determinant, then the inverse of the matrix is computed as $\mathbf{A}^{-1}[i][j] = (1)^{i+j} (D_{ij}) / \det(\mathbf{A})$, where (D_{ij}) is the sub-determinant formed by deleting the i th row and the j th column of $\mathbf{A}$ and $\det(\mathbf{A})$ is the determinant of

$\mathbf{A}$. For our purposes, all arithmetic is done mod 26.

In general terms, the Hill system can be expressed as follows:

$$\mathbf{C} = \mathbf{E}(\mathbf{K}, \mathbf{P}) = \mathbf{K}\mathbf{P} \text{ mod } 26$$

$$\mathbf{P} = \mathbf{D}(\mathbf{K}, \mathbf{P}) = \mathbf{K}^{-1}\mathbf{C} \text{ mod } 26 = \mathbf{K}^{-1}\mathbf{K}\mathbf{P} = \mathbf{P}$$

As with Playfair, the strength of the Hill cipher is that it completely hides single-letter frequencies. Indeed, with Hill, the use of a larger matrix hides more frequency information. Thus a 3×3 Hill cipher hides not only single-letter but also two-letter frequency information.

Although the Hill cipher is strong against a ciphertext-only attack, it is easily broken with a known plaintext attack. For an $m \times m$ Hill cipher, suppose we have m plaintext-ciphertext pairs, each of length m . We label the pairs

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Although the Hill cipher is strong against a ciphertext-only attack, it is easily broken with a known plaintext attack. For an $m \times m$ Hill cipher, suppose we have m plaintext-ciphertext pairs, each of length m . We label the pairs

$$\mathbf{P}_j = \begin{pmatrix} p_{1j} \\ p_{2j} \\ \vdots \\ p_{mj} \end{pmatrix} \text{ and } \mathbf{C}_j = \begin{pmatrix} c_{1j} \\ c_{2j} \\ \vdots \\ c_{mj} \end{pmatrix} \text{ such that } \mathbf{C}_j = \mathbf{K}\mathbf{P}_j \text{ for } 1 \leq j \leq m \text{ and for some}$$

unknown key matrix $\mathbf{K}$. Now define two $m \times m$ matrices $\mathbf{X} = (\mathbf{P}_{ij})$ and $\mathbf{Y} = (\mathbf{C}_{ij})$. Then we can form the matrix equation $\mathbf{Y} = \mathbf{K}\mathbf{X}$. If $\mathbf{X}$ has an inverse, then we can determine

$$\mathbf{K} = \mathbf{Y}\mathbf{X}^{-1}$$

If $\mathbf{X}$ is not invertible, then a new version of $\mathbf{X}$ can be formed with additional plaintext-ciphertext pairs until an invertible $\mathbf{X}$ is obtained.

We use an example. Suppose that the plaintext "friday" is encrypted using a 2 x 2 Hill cipher to yield the

ciphertext PQCFKU. Thus, we know that

$$\mathbf{K} \begin{pmatrix} 5 \\ 17 \end{pmatrix} \bmod 26 = \begin{pmatrix} 15 \\ 16 \end{pmatrix}; \mathbf{K} \begin{pmatrix} 8 \\ 3 \end{pmatrix} \bmod 26 = \begin{pmatrix} 2 \\ 5 \end{pmatrix}; \quad \text{and} \quad \mathbf{K} \begin{pmatrix} 0 \\ 24 \end{pmatrix} \bmod 26 = \begin{pmatrix} 10 \\ 20 \end{pmatrix}$$

Using the first two plaintext-ciphertext pairs, we have

$$\begin{pmatrix} 15 & 2 \\ 16 & 5 \end{pmatrix} = \mathbf{K} \begin{pmatrix} 5 & 8 \\ 17 & 3 \end{pmatrix} \bmod 26$$

The inverse of $\mathbf{X}$ can be computed:

$$\begin{pmatrix} 5 & 8 \\ 17 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 9 & 2 \\ 1 & 15 \end{pmatrix}$$

So

$$\mathbf{K} = \begin{pmatrix} 15 & 2 \\ 16 & 5 \end{pmatrix} \begin{pmatrix} 9 & 2 \\ 1 & 15 \end{pmatrix} = \begin{pmatrix} 137 & 60 \\ 149 & 107 \end{pmatrix} \bmod 26 = \begin{pmatrix} 7 & 8 \\ 19 & 3 \end{pmatrix}$$

This result is verified by testing the remaining plaintext-ciphertext pair.

Lec. 4

5-Affine cipher

Affine cipher is a classical cipher, which is based on a certain equation according to two key. The key are different length, first one has inverse number for decryption equation. A second one is smaller than first one.

Encryption function:

$$Y = a \cdot x + b \pmod{26} \dots\dots\dots (1)$$

The possible values that a could be are 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, and 25.

Table (1): inverse of a											
1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

Decryption function :

$$X = \text{inv}(a) (y - b) \pmod{26} \dots\dots\dots (2)$$

Example: the plaintext to be encrypted is "AFFINE CIPHER" $a=5$ $b=8$

plaintext:	A	F	F	I	N	E	C	I	P	H	E	R
x:	0	5	5	8	13	4	2	8	15	7	4	17
5x+8	8	33	33	48	73	28	18	48	83	43	28	93
(5x+8)mod 26	8	7	7	22	21	2	18	22	5	17	2	15
cipher text:	I	H	H	W	V	C	S	W	F	R	C	P

The cipher to decryption:

ciphertext:	I	H	H	W	V	C	S	W	F	R	C	P
y:	8	7	7	22	21	2	18	22	5	17	2	15
21(y-8):	0	-21	-21	294	273	-126	210	294	-63	189	-126	147
(21(y-8)) mod 26:	0	5	5	8	13	4	2	8	15	7	4	17
plaintext:	A	F	F	I	N	E	C	I	P	H	E	R

6) Transposition:

means a management of letters according to some schema. i.e. rearrange bits or characters in the data.

1. The plaintext was written into the figure according to some (written) path.
2. The cipher text was taken off the figure according to some (take off) path.
3. The key consisted of the figure together with the written in & take off path.

Transposition Types:

1) Simple transposition (columnar transposition):

The plaintext can be written into a matrix by rows the cipher text can obtain by taking the columns in some order.

Example 1 :

CRYPTOGRAPHY = plain text

3 1 4 2 = key

(Encipherment process)

(Encipherment process)							
3	1	4	2				
C	R	Y	P				
T	O	G	R	+ key	→	ROP	PRY
A	P	H	Y			CTA	YGH
						1	2
						3	4

(Decipherment process)			
key = 3 1 4 2			
cipher text = ROP PRY CTA YGH			
	1	2	3
			4

3	1	4	2				
C	R	Y	P				
T	O	G	R	→	C	R	Y
A	P	H	Y		P	R	Y

Example 2:

Plaintext = this is transposition

Key = code

A	B	¹ C	² D	³ E	F	G	H	I	J	K	L	M	N	⁴ O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Code=1 4 2 3

(Encipherment process)

1	4	2	3
T	h	i	s
I	s	t	r
A	n	s	p
O	s	i	t
I	o	n	x

cipher text = tiaoi itsin srptx hsnso

(Decipherment process)

cipher text = tiaoi itsin srptx hsnso

1 2 3 4

key= code = 1 4 2 3

1	4	2	3
t	h	i	s
i	s	t	r
a	n	s	p
o	s	i	t
i	o	n	X

2) A fixed period d with a permutation function.

$F: \mathbb{Z}_d \rightarrow \mathbb{Z}_d$

Example :

Plain text: CRYPTOGRAPHY , d=4 , f=2 4 1 3

(Encipherment process)

d=4	d=4	d=4
CRYP	TOGR	APHY
1 2 3 4	1 2 3 4	1 2 3 4
Cipher text : RPCY	ORTG	PYAH

(Decipherment process)

Cipher text : RPCY ORTG PYAH , d=4, f= 2413

2 4 1 3	2413	2413
Plaintext : CRYP	TOGR	APHY
1 2 3 4	12 3 4	1 2 3 4

Note: d represent block length

a) Simple Substitution

replace each plaintext letter with a corresponding cipher text letter.

1- Keyword

Example:

Keyword= CRYPTOGRAPHIC SYSTEM

P: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

K: C R Y P T O G A H I S E M B D F J K L N Q U V W X Z

To encipher the plaintext= CRYPTOGRAPHY

ciphertext=YKXFNDGKCFAX

2- Shift alphabets

$F(a)=(a+k)\bmod n$ encipher

$M=(c-k)\bmod n$ decipher

i.e. $c=(p+k)\bmod 26$, where $a =0,.....z=25$

- this is called shifted alphabey by k position

- example (caesar cipher use $k=3$)

P= a b c d.....z

C= d e f g.....c

3-Multiplication

$$F(a)=(a.k)\bmod n$$

Or

$$C=(P.k)\bmod 26, \text{ where } \text{GCD}(k,26)=1$$

Example:

$$K=9$$

P: A B C D E F G H I J KZ

C: A J S B K T C L U D.....R

Lec. 5

7- Playfair Cipher

The most common definition for playfair cipher is it operates on pairs of letters. The cipher text is broken into blocks, each block contains two letters. The key is 5x5 square consisting of every letter except j.

The **Playfair cipher** was the first practical digraph substitution cipher. The scheme was invented in **1854** by **Charles Wheatstone** but was named after Lord Playfair who promoted the use of the cipher. In playfair cipher unlike [traditional cipher](#) we encrypt a pair of alphabets(digraphs) instead of a single alphabet.

It was used for tactical purposes by British forces in the Second Boer War and in World War I and for the same purpose by the Australians during World War II. This was because Playfair is reasonably fast to use and requires no special equipment

Encryption Technique

The algorithm consists of 2 steps:

1. **Generate the key Square (5x5):**

- The key square is a 5×5 grid of alphabets that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). If the plaintext contains J, then it is replaced by I.
- The initial alphabets in the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters of the alphabet in order.

2. **Algorithm to encrypt the plain text:** The plaintext is split into pairs of two letters (digraphs). If there is an odd number of letters, a Z is added to the last letter.

Example

PlainText: "instruments"

After Split: 'in' 'st' 'ru' 'me' 'nt' 'sz'

Explanation: Pair cannot be made with same letter. Break the letter in single and add a bogus letter to the previous letter. Here 'z' is the bogus letter.

Plain Text: "hello"

After Split: 'he' 'lx' 'lo'

Explanation: Here 'x' is the bogus letter.

Plain Text: "heloe"

After Split: 'he' 'lx' 'lo' 'ez'

Explanation: If the letter is standing alone in the process of pairing, then add an extra bogus letter with the alone letter. Here 'x' and 'z' are the bogus letters.

Rules for Encryption

If both the letters are in the same column: Take the letter below each one (going back to the top if at the bottom).

For example:

Diagraph: "me"

Encrypted Text: cl

Encryption: m -> c e -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

If both the letters are in the same row: Take the letter to the right of each one (going back to the leftmost if at the rightmost position).

For example:

Diagraph: "st"

Encrypted Text: tl

Encryption: s -> t t -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

If neither of the above rules is true: Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.

For example:

Diagraph: "nt"

Encrypted Text: rq

Encryption: n -> r t -> q

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

For example:

Plain ext : “instrumentsz”

Encrypted text: gatlmzclrqtx

gatlmzclrqtx

in:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

st:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

ru:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

me:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

nt:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

sz:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

C++

```
#include <bits/stdc++.h>
```

```
using namespace std;
```

```
// Function to convert the string to lowercase
```

```
void toLowerCase(string &plain) {
```

```
    int n = plain.size();
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (plain[i] > 64 && plain[i] < 91)
```

```
            plain[i] += 32;
```

```
    }
```

```
}
```

```
// Function to remove all spaces in a string
```

```
void removeSpaces(string &plain) {
```

```
    int n = plain.size();
```

```
    string temp;
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (plain[i] != ' ') {
```

```
            temp += plain[i];
```

```
        }
```

```
    }
```

```
    plain = temp;
```

```
}
```

// Function to generate the 5x5 key square

```
void generateKeyTable(string &key,
    vector<vector<char>> &keyT) {
    int n = key.size();
    // 5x5 key table
    keyT.resize(5, vector<char>(5, 0));
    // a 26 character hashmap
    // to store count of the alphabet
    vector<int> hash(26, 0);
    int i, j, k, flag = 0;
    for (i = 0; i < n; i++) {
        if (key[i] != 'j')
            hash[key[i] - 97] = 2;
    }
    hash['j' - 97] = 1;
    i = 0;
    j = 0;
    for (k = 0; k < n; k++) {
        if (hash[key[k] - 97] == 2) {
            hash[key[k] - 97] -= 1;
            keyT[i][j] = key[k];
            j++;
            if (j == 5) {
                i++;
                j = 0;
            }
        }
    }
}
```

```
for (k = 0; k < 26; k++) {  
    if (hash[k] == 0) {  
        keyT[i][j] = (char)(k + 97);  
        j++;  
        if (j == 5) {  
            i++;  
            j = 0;  
        }  
    }  
}  
}
```

// Function to search for the characters of a digraph

// in the key square and return their position

```
void search(vector<vector<char>> &keyT,  
            char a, char b, vector<int> &arr) {  
    int i, j;  
    if (a == 'j')  
        a = 'i';  
    else if (b == 'j')  
        b = 'i';  
    for (i = 0; i < 5; i++) {  
        for (j = 0; j < 5; j++) {  
            if (keyT[i][j] == a) {  
                arr[0] = i;  
                arr[1] = j;  
            }  
            else if (keyT[i][j] == b) {  
                arr[2] = i;
```

```
        arr[3] = j;
    }
}
}
}

// Function to make the plain text length to be even
int prepare(string &str) {
    if (str.size() % 2 != 0) {
        str += 'z';
    }
    int n = str.size();
    return n;
}

// Function for performing the encryption
void encrypt(string &str, vector<vector<char>> &keyT) {
    int n = str.size();
    vector<int> arr(4);
    for (int i = 0; i < n; i += 2) {
        search(keyT, str[i], str[i + 1], arr);
        if (arr[0] == arr[2]) {
            str[i] = keyT[arr[0]][(arr[1] + 1) % 5];
            str[i + 1] = keyT[arr[0]][(arr[3] + 1) % 5];
        }
        else if (arr[1] == arr[3]) {
            str[i] = keyT[(arr[0] + 1) % 5][arr[1]];
            str[i + 1] = keyT[(arr[2] + 1) % 5][arr[1]];
        }
        else {
            str[i] = keyT[arr[0]][arr[3]];

```

```
        str[i + 1] = keyT[arr[2]][arr[1]];
    }
}
}

// Function to encrypt using Playfair Cipher
void encryptByPlayfairCipher(string &str, string &key) {
    vector<vector<char>> keyT;
    removeSpaces(key);
    toLowerCase(key);
    toLowerCase(str);
    removeSpaces(str);
    prepare(str);
    generateKeyTable(key, keyT);
    encrypt(str, keyT);
}

int main() {
    string key = "Monarchy";
    string str = "instruments";
    cout << "Key text: " << key << endl;
    cout << "Plain text: " << str << endl;
    encryptByPlayfairCipher(str, key);
    cout << "Cipher text: " << str << endl;
    return 0;
}
```

in python

Function to convert the string to lowercase

```
def toLowerCase(plain):
```

```
    n = len(plain)
```

```
    result = ""
```

```
    for i in range(n):
```

```
        if 64 < ord(plain[i]) < 91:
```

```
            result += chr(ord(plain[i]) + 32)
```

```
        else:
```

```
            result += plain[i]
```

```
    return result
```

Function to remove all spaces in a string

```
def removeSpaces(plain):
```

```
    n = len(plain)
```

```
    temp = ""
```

```
    for i in range(n):
```

```
        if plain[i] != ' ':
```

```
            temp += plain[i]
```

```
    return temp
```

Function to generate the 5x5 key square

```
def generateKeyTable(key, keyT):
```

```
    n = len(key)
```

```
    keyT.clear()
```

```
    for i in range(5):
```

```
        keyT.append([0]*5)
```

```
    hashMap = [0]*26
```

```
    for i in range(n):
```

```
        if key[i] != 'j':
```

```
hashMap[ord(key[i]) - 97] = 2
```

```
hashMap[ord('j') - 97] = 1
```

```
i = 0
```

```
j = 0
```

```
for k in range(n):
```

```
    if hashMap[ord(key[k]) - 97] == 2:
```

```
        hashMap[ord(key[k]) - 97] -= 1
```

```
        keyT[i][j] = key[k]
```

```
        j += 1
```

```
    if j == 5:
```

```
        i += 1
```

```
        j = 0
```

```
for k in range(26):
```

```
    if hashMap[k] == 0:
```

```
        keyT[i][j] = chr(k + 97)
```

```
        j += 1
```

```
    if j == 5:
```

```
        i += 1
```

```
        j = 0
```

```
# Function to search for the characters of a digraph
```

```
# in the key square and return their position
```

```
def search(keyT, a, b, arr):
```

```
if a == 'j':
```

```
    a = 'i'
```

```
if b == 'j':
```

```
    b = 'i'
```

```
for i in range(5):
```

```
    for j in range(5):
```

```
        if keyT[i][j] == a:
```

```
            arr[0] = i
```

```
            arr[1] = j
```

```
        elif keyT[i][j] == b:
```

```
            arr[2] = i
```

```
            arr[3] = j
```

```
# Function to make the plain text length to be even
```

```
def prepare(string):
```

```
    if len(string) % 2 != 0:
```

```
        string += 'z'
```

```
    return string
```

```
# Function for performing the encryption
```

```
def encrypt(string, keyT):
```

```
    n = len(string)
```

```
    arr = [0]*4
```

```
    result = list(string)
```

```
    for i in range(0, n, 2):
```

```
search(keyT, result[i], result[i+1], arr)

if arr[0] == arr[2]:

    result[i] = keyT[arr[0]][(arr[1] + 1) % 5]

    result[i+1] = keyT[arr[0]][(arr[3] + 1) % 5]

elif arr[1] == arr[3]:

    result[i] = keyT[(arr[0] + 1) % 5][arr[1]]

    result[i+1] = keyT[(arr[2] + 1) % 5][arr[1]]

else:

    result[i] = keyT[arr[0]][arr[3]]

    result[i+1] = keyT[arr[2]][arr[1]]

return ".join(result)

# Function to encrypt using Playfair Cipher

def encryptByPlayfairCipher(string, key):

    keyT = []

    key = toLowerCase(removeSpaces(key))

    string = toLowerCase(removeSpaces(string))

    string = prepare(string)

    generateKeyTable(key, keyT)

    return encrypt(string, keyT)

key = "Monarchy"

string = "instruments"

print("Key text:", key)

print("Plain text:", string)
```

```
string = encryptByPlayfairCipher(string, key)
```

```
print("Cipher text:", string)
```

Output

```
Key text: Monarchy  
Plain text: instruments  
Cipher text: gatlmzclrqtx
```

Decryption Technique

Decrypting the Playfair cipher is as simple as doing the same process in reverse. The receiver has the same key and can create the same key table, and then decrypt any messages made using that key.

The Algorithm consists of 2 steps:

1. **Generate the key Square(5x5) at the receiver's end:**
 - The key square is a 5×5 grid of alphabets that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). If the plaintext contains J, then it is replaced by I.
 - The initial alphabets in the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters of the alphabet in order.
2. **Algorithm to decrypt the ciphertext:** The ciphertext is split into pairs of two letters (digraphs).

Note: The **ciphertext** always have **even** number of characters.

Rules for Decryption

If both the letters are in the same column: Take the letter above each one (going back to the bottom if at the top).

For example:

Diagraph: "cl"

Decrypted Text: me

Decryption: c -> m l -> e

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

If both the letters are in the same row: Take the letter to the left of each one (going back to the rightmost if at the leftmost position).

For example:

Diagraph: "tl"

Decrypted Text: st

Decryption: t -> s l -> t

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

f neither of the above rules is true: Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.

For example:

Diagraph: "rq"

Decrypted Text: nt

Decryption: r -> n q -> t

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

or example:

Plain Text: "gatlmzclrqtx"

Decrypted Text: instrumentsz

Decryption:

(red)-> (green)

ga -> in

tl -> st

mz -> ru

cl -> me

rq -> nt

tx -> sz

in:	<table> <tr><td>M</td><td>O</td><td>N</td><td>A</td><td>R</td></tr> <tr><td>C</td><td>H</td><td>Y</td><td>B</td><td>D</td></tr> <tr><td>E</td><td>F</td><td>G</td><td>I</td><td>K</td></tr> <tr><td>L</td><td>P</td><td>Q</td><td>S</td><td>T</td></tr> <tr><td>U</td><td>V</td><td>W</td><td>X</td><td>Z</td></tr> </table>	M	O	N	A	R	C	H	Y	B	D	E	F	G	I	K	L	P	Q	S	T	U	V	W	X	Z	st:	<table> <tr><td>M</td><td>O</td><td>N</td><td>A</td><td>R</td></tr> <tr><td>C</td><td>H</td><td>Y</td><td>B</td><td>D</td></tr> <tr><td>E</td><td>F</td><td>G</td><td>I</td><td>K</td></tr> <tr><td>L</td><td>P</td><td>Q</td><td>S</td><td>T</td></tr> <tr><td>U</td><td>V</td><td>W</td><td>X</td><td>Z</td></tr> </table>	M	O	N	A	R	C	H	Y	B	D	E	F	G	I	K	L	P	Q	S	T	U	V	W	X	Z	ru:	<table> <tr><td>M</td><td>O</td><td>N</td><td>A</td><td>R</td></tr> <tr><td>C</td><td>H</td><td>Y</td><td>B</td><td>D</td></tr> <tr><td>E</td><td>F</td><td>G</td><td>I</td><td>K</td></tr> <tr><td>L</td><td>P</td><td>Q</td><td>S</td><td>T</td></tr> <tr><td>U</td><td>V</td><td>W</td><td>X</td><td>Z</td></tr> </table>	M	O	N	A	R	C	H	Y	B	D	E	F	G	I	K	L	P	Q	S	T	U	V	W	X	Z
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Advantages:

- It is significantly harder to break since the frequency analysis technique used to break simple substitution ciphers is difficult but still can be used on $(25 \times 25) = 625$ digraphs rather than 25 monographs which is difficult.
- Frequency analysis thus requires more cipher text to crack the encryption.

Disadvantages:

- An interesting weakness is the fact that a digraph in the ciphertext (AB) and it's reverse (BA) will have corresponding plaintexts like UR and RU (and also ciphertext UR and RU will correspond to plaintext AB and BA, i.e. the substitution is self-inverse). That can easily be exploited with the aid of frequency analysis, if the language of the plaintext is known.
 - Another disadvantage is that playfair cipher is a [symmetric cipher](#) thus same key is used for both encryption and decryption.

Lec. 6

8- One-Time Pad

*Use a random key that is as long as the message, so that the key need not be repeated.

*the key is to be used to encrypt and decrypt a single message, and then is discarded.

*Each new message requires a new key of the same length as the new message.

* one-time pad, is unbreakable.

((It produces random output that bears no statistical relationship to the plaintext)).

* Because the ciphertext provides no information about the original message to a cryptanalyst, there is simply no way to break the code .

ENCRYPTION

EXAMPLE:

H	E	L	L	O	
7(h)	4(e)	11(l)	11(l)	14(o)	← plaintext
+					
23(X)	12(M)	2(C)	10(K)	11(L)	← key
= 30	16	13	21	25	← plaintext + key
=4(E)	16(Q)	13(N)	21(V)	25(Z)	← (p+ key)mod26
E	Q	N	V	Z	← cipher text

DECRYPTION

EXAMPLE:

E	Q	N	V	Z	
4(E)	16(Q)	13(N)	21(V)	25(Z)	← cipher text
-					
23(X)	12(M)	2(C)	10(K)	11(L)	← key
	7				
= -19	4	11	11	14	← ciphertext-key
= -19+26	4	11	11	14	← (c-k)mod26
h	e	l	l	o	← plain text

Encrypt the following text “otp” using otp with logic operation key=xyz

Plaintext= otp

Key = xyz

o=ASCII code for letter “o” =111 as decimal number so $111_{10} = (01101111)_2$

x= ASCII code for letter “x”= 120 as decimal number so $120_{10} = (01111000)_2$

o XOR x =

o= $111_{10} = (01101111)_2$

x= $120_{10} = (01111000)_2$

0 1 1 0 1 1 1 1

XOR

0 1 1 1 1 0 0 0

$(00010111)_2 = 23_{10} \mod 26 = 23 = x$ o xor x =X =23+65=88=X

t XOR y =

t= $116_{10} = (01110100)_2$

y= $121_{10} = (01111001)_2$

0 1 1 1 0 1 0 0

XOR

0 1 1 1 1 0 0 1

$(00001101)_2 = 13_{10} \mod 26 = 13 = n$ t XOR y =N =13+65=78

p XOR z =

p= $112_{10} = (01110000)_2$

z= $122_{10} = (01111010)_2$

0 1 1 1 0 0 0 0

XOR

0 1 1 1 1 0 1 0

$(00001010)_2 = 10_{10} \mod 26 = 10 = k$ p XOR z = K=10+65=75

Q2: Decrypt the following cipher text "XNK" using otp with logic operation key=xyz

Cipher text= XNK

Key = xyz

X=ASCII code for letter "X" =88 as decimal number so $88-65=23_{10}=00010111_2$

x= ASCII code for letter "x"= 120 as decimal number so $120_{10}=(01111000)_2$

X XOR x =

X= $88_{10}=88 - \text{ascii for capital letter (65)} = 23=(00010111)_2$

x= $120_{10}=(01111000)_2$

0 0 0 1 0 1 1 1

XOR

0 1 1 1 1 0 0 0

(0 1 1 0 1 1 1 1)₂ = 111_{10} = ascii code 111 =letter o

N XOR y =

78-65=13

n= $13_{10}=(00001101)_2$

y= $121_{10}=(01111001)_2$

0 0 0 0 1 1 0 1

XOR

0 1 1 1 1 0 0 1

(0 1 1 1 0 1 0 0)₂ = 116_{10} =t n XOR y =t

k XOR z =

K=75-65=10

k= $10_{10}=(00001010)_2$

z= $122_{10}=(01111010)_2$

0 0 0 0 1 0 1 0

XOR

0 1 1 1 1 0 1 0

(0 1 1 1 0 0 0 0)₂ = 112_{10} =p K XOR z = p

(XNK) XOR (xyz)= (otp)

Cryptanalysis

In this section, we discuss some techniques of cryptanalysis. The general assumption that is usually made is that the opponent, Oscar, knows the cryptosystem being used. This is usually referred to as *Kerckhoff's principle*. Of course, if Oscar does not know the cryptosystem being used, that will make his task more difficult. But we do not want to base the security of a cryptosystem on the (possibly shaky) premise that Oscar does not know what system is being employed. Hence, our goal in designing a cryptosystem will be to obtain security under Kerckhoff's principle.

First, we want to differentiate between different levels of attacks on cryptosystems. The most common types are enumerated as follows.

Ciphertext-only

The opponent possesses a string of ciphertext, y .

Known plaintext

The opponent possesses a string of plaintext, x , and the corresponding ciphertext, y .

Chosen plaintext

The opponent has obtained temporary access to the encryption machinery. Hence he can choose a plaintext string, x , and construct the corresponding ciphertext string, y .

Chosen ciphertext

Chosen ciphertext

The opponent has obtained temporary access to the decryption machinery. Hence he can choose a ciphertext string, y , and construct the corresponding plaintext string, x .

In each case, the object is to determine the key that was used. We note that a chosen ciphertext attack is relevant to public-key cryptosystems, which we discuss in the later chapters.

We first consider the weakest type of attack, namely a ciphertext-only attack. We also assume that the plaintext string is ordinary English text, without punctuation or “spaces.” (This makes cryptanalysis more difficult than if punctuation and spaces were encrypted.)

Many techniques of cryptanalysis use statistical properties of the English language. Various people have estimated the relative frequencies of the 26 letters by compiling statistics from numerous novels, magazines, and newspapers. The estimates in Table 1.1 were obtained by Beker and Piper.

On the basis of the above probabilities, Beker and Piper partition the 26 letters into five groups as follows:

1. *E*, having probability about 0.120
2. *T, A, O, I, N, S, H, R*, each having probabilities between 0.06 and 0.09
3. *D, L*, each having probabilities around 0.04
4. *C, U, M, W, F, G, Y, P, B*, each having probabilities between 0.015 and 0.028

It may also be useful to consider sequences of two or three consecutive letters called *digrams* and *trigrams*, respectively. The 30 most common digrams are (in decreasing order) *TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI*, and *OF*. The twelve most common trigrams are (in decreasing order) *THE ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR*, and *DTH*.

Table 1.1 Probabilities of Occurrence of the 26 Letters

letter	probability	letter	probability
<i>A</i>	.082	<i>N</i>	.067
<i>B</i>	.015	<i>O</i>	.075
<i>C</i>	.028	<i>P</i>	.019
<i>D</i>	.043	<i>Q</i>	.001
<i>E</i>	.127	<i>R</i>	.060
<i>F</i>	.022	<i>S</i>	.063
<i>G</i>	.020	<i>T</i>	.091
<i>H</i>	.061	<i>U</i>	.028
<i>I</i>	.070	<i>V</i>	.010
<i>J</i>	.002	<i>W</i>	.023
<i>K</i>	.008	<i>X</i>	.001
<i>L</i>	.040	<i>Y</i>	.020
<i>M</i>	.024	<i>Z</i>	.001

Cryptanalysis Monoalphabetic Ciphers

Cryptanalysis is the science and study of methods of breaking ciphers. It is assumed that the ciphertext is sent over insecure communications lines and is available to the cryptanalyst. His aim is to recover the plaintext from the ciphertext without knowing the key parameters[17][11][18]. A cipher is breakable if it is possible to determine the plaintext or key parameters from the ciphertext. There are three basic methods of attack : ciphertext-only, known-plaintext, and chosen-plaintext. Under a ciphertext-only attack, a cryptanalyst must determine the key solely from intercepted ciphertext, though the method of encryption and certain probable words may be known. On the other hand, a knownplaintext attack requires a substantial amount of plaintext and ciphertext be known, and a chosen-plaintext attack is able to acquire the ciphertext corresponding to selected plaintext[21, 25].

2-5-1 Cryptanalysis a mono alphabetic substitution ciphers

If the cryptanalysis knows the nature of the plaintext (e.g., non compressed English text), then the analyst can exploit the regularities of the language. To see how such a cryptanalysis might proceed, we give a partial example here that is adapted from one in. The cipher text to be solved is:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ

VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX

EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

As a first step, the relative frequency of the letters can be determine and compare to a standard frequency distribution for English, such as is shown in fig. 3.3. If the message were long enough, this technique alone might be sufficient, but because this is a relatively short message, we cannot expect an exact match. In any case, the relative frequencies of the letters in the cipher text (in percentages) are as follows:-

P 13.33	H 5.83	F 3.33	B 1.67	C 0.000
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.000
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.000
U 8.33	V 4.17	T 2.50	I 0.83	N 0.000
O 7.50	X 4.17	A 1.67	J 0.83	R 0.000
M 6.67				

Comparing this breakdown with figure 3.3 , it seems likely that cipher letters P and Z are equivalents of plain letters e and t, but it is not certain which is which. The letters S, U, O, M, and H are all of relatively high frequency and probably correspond to plain letters from the set {a, h, i, n o, r, s}. The letters with the lowest frequencies (namely, A, B, G, Y, I, J) are likely included in the set The genetic algorithms are.

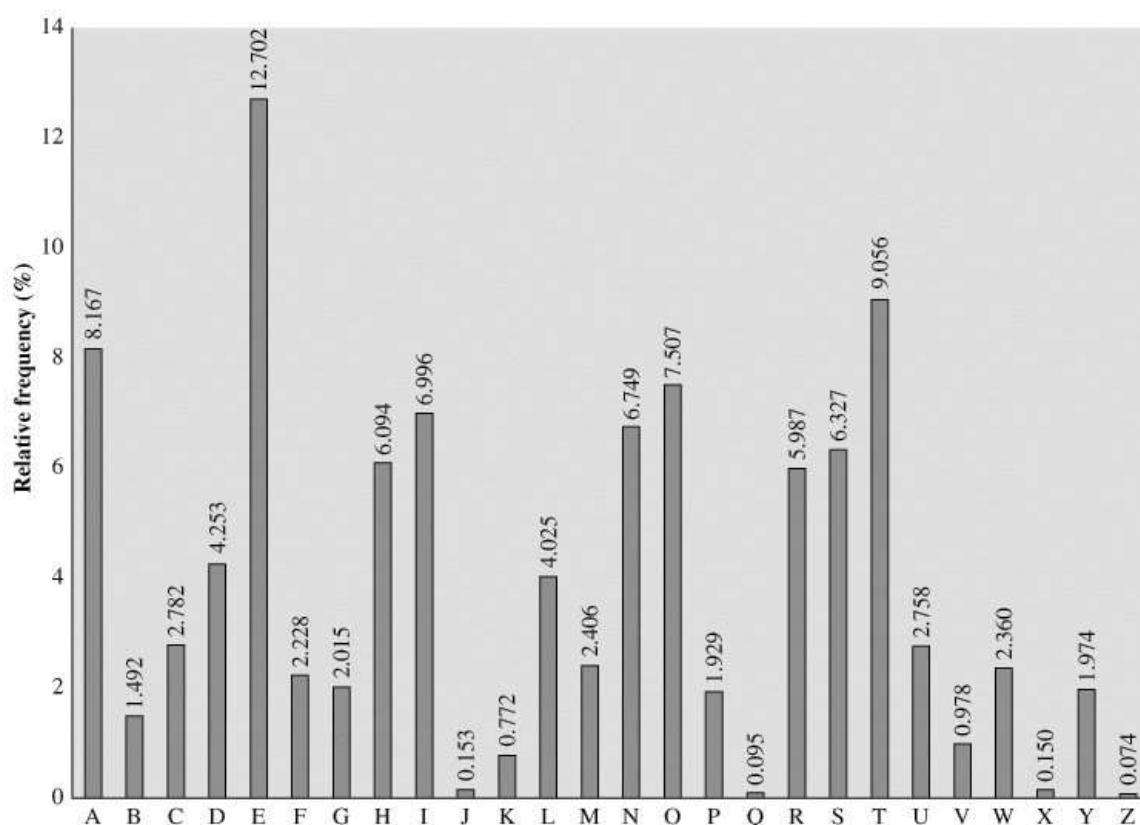


Figure.3.3: Letter Frequency for English Language

There are a number of ways to proceed at this point. We could make some tentative assignments and start to fill in the plaintext to see if it looks like a reasonable "skeleton" of a message. A more systematic approach is to look for other regularities. For example, certain words may be known to be in the text. Or we could look for repeating sequences of chapter letters and try to deduce their plaintext equivalents.

A powerful tool is to look at the frequency of two-letter combinations, known as digrams. A table similar to figure 3.3 could drawn up showing the relative frequency of digrams. The most common such digram is th. In our ciphertext, the most common digram is ZW, which appears three times. So we make the correspondence of Z with t and W with h. Then, by our earlier hypothesis , we can equate P with e. Now notice that the sequence ZWP appears in the ciphertext, and we can translate that sequence as "the".

U	Z	Q	S	O	V	U	O	H	X	M	O	P	V	G	P	O	Z	P	E	V	S	G	Z	W	S	Z	O	P	F
	t		a									e			e		t	e			a		t	h	a	t		e	
P	E	S	X	U	D	B	M	E	T	S	X	A	I	Z	V	U	E	P	H	Z	H	M	D	Z	S	H	Z	O	W
e		a								a								e		t				t	a		t		h
S	F	P	A	P	P	D	T	S	V	P	Q	U	Z	W	Y	M	X	U	Z	U	H	S	X	E	P	Y	E	P	O
a		e		e	e			a		e			t	h					t			a			e			e	
P	D	Z	S	Z	U	F	P	O	M	B	Z	W	P	F	U	P	Z	H	M	D	J	U	D	T	M	O	H	M	Q
e		t	a	t			e				t	h	e				t												

Only four letters have been identified, but already we have quite a bit of the message. Continued analysis of frequencies plus trial and error should easily yield a solution from this point. The complete plaintext, with spaces added between words, follows:

**it was disclosed yesterday that several informal but
direct contacts have been made with political**

Mono alphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet

it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow

Table 1: Types of attacks on encrypted messages

Type of Attack	Known to Cryptanalyst
Cipher text Only	• Encryption algorithm • Cipher text
Known Plaintext	• Encryption algorithm • Cipher text
	• One or more plaintext–cipher text pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Cipher text • Plain text message chosen by cryptanalyst, together with its corresponding cipher text generated with the secret key
Chosen Cipher text	• Encryption algorithm • Cipher text • Cipher text chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Cipher ext • Plaintext message chosen by cryptanalyst, together with its corresponding Cipher text generated with the secret key • Cipher text chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

(1.8) Cryptanalysis:

The science of deducing the plaintext from a ciphertext, without knowledge of the key.

(1.8.1) Attacks on encrypted messages

The objective of the following attacks is to systematically recover plaintext from ciphertext, or even more drastically, to deduce the decryption key.

1. **A ciphertext-only attack:** is one where the adversary (or cryptanalyst) tries to deduce the decryption key or plaintext by only observing ciphertext. Any encryption scheme vulnerable to this type of attack is considered to be completely insecure.
2. **A known-plaintext attack:** is one where the adversary has a quantity of plaintext and corresponding ciphertext. This type of attack is typically only marginally more difficult to mount.
3. **A chosen-plaintext attack:** is one where the adversary chooses plaintext and is then given corresponding ciphertext. Subsequently, the adversary uses any information deduced in order to recover plaintext corresponding to previously unseen ciphertext.
4. **An adaptive chosen-plaintext attack:** is a chosen-plaintext attack wherein the choice of plaintext may depend on the ciphertext received from previous requests.
5. **A chosen-ciphertext attack:** is one where the adversary selects the ciphertext and is then given the corresponding plaintext. One way to mount such an attack is for the adversary to gain access to the equipment used for decryption (but not the decryption key, which may be securely embedded in the equipment). The objective is then to be able, without access to such equipment, to deduce the plaintext from (different) ciphertext.
6. **An adaptive chosen-ciphertext attack:** is a chosen-ciphertext attack where the choice of ciphertext may depend on the plaintext received from previous requests.

(1.8.2) Some concepts on cryptanalysis:

- ✦ **Frequency:** number of appearance of the letter in the ciphertext, where the frequencies of the ciphertext letters are compared with the frequencies in Table 1 or Figure 5.
- ✦ **Repetition:** is the similar parts in the ciphertext that have length not less than three. This helps us to find the length of the key (the number of alphabets that used to enciphering in the polyalphabetic systems).
Take the Highest Common Factor HCF between the repetitions, which represent the length of the key, this method, is called the Kasiski method.

Letter	%	Letter	%
a	8.167	n	6.749
b	1.492	o	7.507
c	2.782	p	1.929
d	4.253	q	0.095
e	12.702	r	5.987
f	2.228	s	6.327
g	2.015	t	9.056
h	6.094	u	2.758
i	6.966	v	0.978
j	0.153	w	2.360
k	0.772	x	0.150
l	4.025	y	1.974
m	2.406	z	0.074

Table 1: English letters frequencies

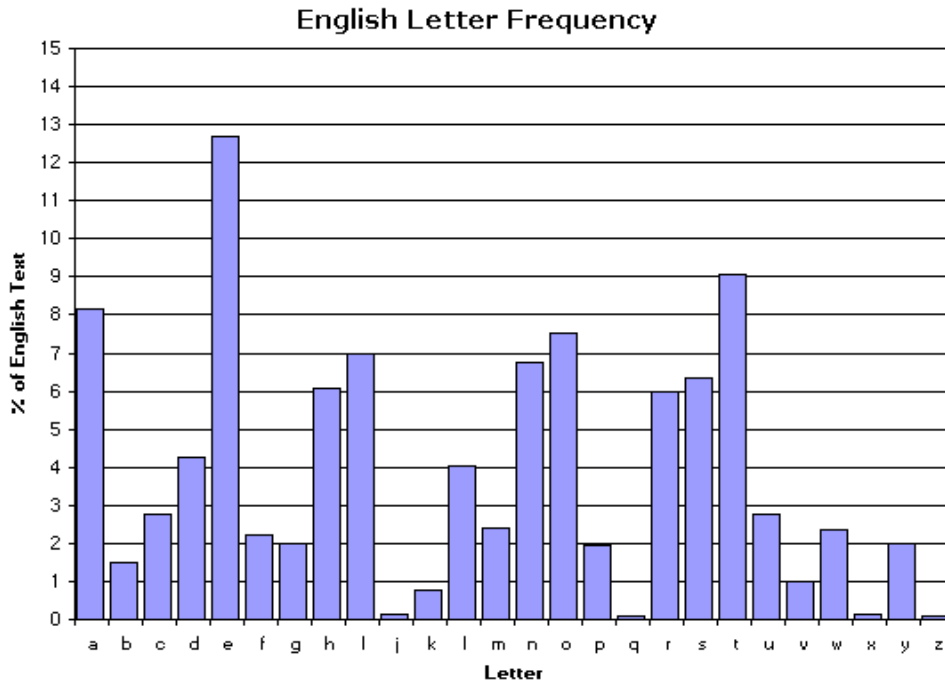


Figure 5: Histogram of English letters frequencies

✦ **Index of Coincidence (IC):** is the probability that two letters selected from the text are identical, we can compute the IC from the following equation:

$$IC = \frac{\sum_{\lambda} f_{\lambda}(f_{\lambda} - 1)}{n(n - 1)},$$

where f_{λ} is the frequency of the letter λ in the ciphertext and n is the length of the letter. The IC value differs from language to another. We can use the IC to discover if the message were enciphered using Monoalphabetic system or polyalphabetic system.

- ✦ **Coincidence:** is the computing of the coincidence of the ciphertexts, where two messages put one over the other, and the purpose is to discover if the two messages were enciphered using the same key. If there is 7 coincidence letters between 100 letters in the two messages then the two messages were enciphered using the same key, while if there is 4 letters coincidence between every 100 letters then they enciphered with different keys.

(1.9) Cryptanalysis examples:

First of all we must specified the type of the cipher system that was used. If the frequencies of the ciphertext are the same as the frequencies of the language then, a transposition cipher system was used; otherwise a substitution cipher system was used.

(1.9.1) Cryptanalysis of transposition cipher systems:

When we decide that a transposition cipher system were used, we put the cipher text in $m \times n$ matrix, m and n depends on the length of the received ciphertext, for example if the length is 500 then one of the possible sizes is 20×25 . Then we rearrange the columns to get some known patterns such as (and, the, ion, that,...) in addition to some expected word in the message.

As we know there are two types of transposition cipher system: simple and double transposition, the cryptanalysis of the last one is more complicated because we lose the ability to find the known patterns.

(1.9.2) Cryptanalysis of substitution cipher systems:

If we know that a substitution cipher system was used, the next step is to determine whether a monoalphabetic system or polyalphabetic system was used, by using the IC of the language.

Example: A sample of ordinary English contains the following distribution of letters

Letter	Count	Letter	count
A	141	N	119
B	36	O	132
C	36	P	28
D	103	Q	1
E	188	R	95
F	37	S	64
G	34	T	182
H	102	U	59
I	123	V	13
J	4	W	55
K	18	X	3
L	56	Y	23
M	27	Z	0

What is the probability of selecting an identical pair of letters from this collection? in other word compute the IC.

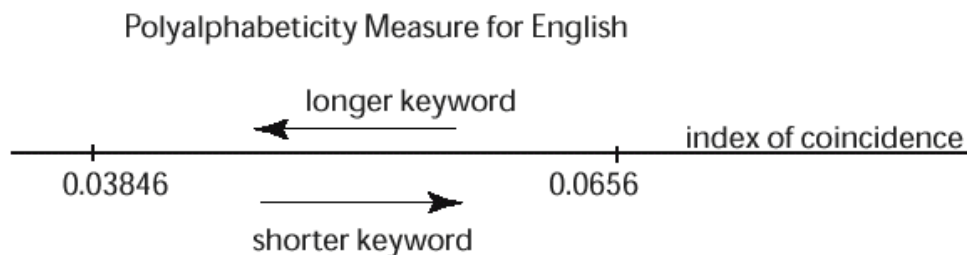
$$IC = \frac{\sum_A f_{\lambda}(f_{\lambda} - 1)}{n(n-1)}$$

$$IC = \frac{141(141-1) + 36(36-1) + \dots + 23(23-1) + 0(0-1)}{1679(1679-1)} = \frac{184838}{2817362} \approx 0.0656.$$

Example: What is the index of coincidence for a collection of 2600 letters consisting of 100 A 's, 100 B 's, 100 C 's, ..., 100 Z 's?

$$IC = \frac{100 \cdot 99 + 100 \cdot 99 + \dots + 100 \cdot 99 + 100 \cdot 99}{2600 \cdot 2599} \approx 0.0384615.$$

As we see from the two examples above the index of coincidence of totally random (uniformly distributed) collection of letters is about 0.0385. Vigenere ciphertexts from longer keywords have a more uniform distribution of letters. For keyword length closer to 1, the index of coincidence will be larger, closer to 0.0656.



If the length of the text is n , we can quantify the connection between index of coincidence and keyword length k , (number of alphabets), where:

$$k \approx \frac{0.0265 \cdot n}{(0.065 - IC) + n(IC - 0.0385)}.$$

Example: A polyalphabetic ciphertext has the following letter counts.

Letter	Count	Letter	count
A	60	N	28
B	50	O	83
C	42	P	44
D	64	Q	69
E	51	R	13
F	63	S	29
G	19	T	66
H	48	U	87
I	56	V	63
J	67	W	19
K	23	X	43
L	45	Y	39
M	44	Z	67

Estimate the keyword length.

Solution: There are $n=1282$ letters.

$$IC = \frac{60 \cdot 59 + 50 \cdot 49 + \dots + 67 \cdot 66}{1282 \cdot 1281} = \frac{35761}{821121} \approx 0.04355.$$

$$K = \frac{0.0265 \cdot 1282}{(0.065 - 0.04355) + 1282(0.04355 - 0.03846)} = 5.1892.$$

Based only on this evidence, a reasonably likely keyword length is 5.

➤ Now, after the above tests if we conclude that a monoalphabetic cipher system was used, then:

❶ If a direct standard or reversed system were used, we compare the frequencies of the ciphertext with the frequencies of the English language, start by putting E against the letter with the higher frequency in the ciphertext, then we put the other letters sequentially.

❷ If a mixed cipher system was used (Random) then we compare the frequencies of the ciphertext with that in Table 1 and Figure 5.

For advanced analysis we can use in addition to Table 1, a table of double letter frequencies TH, HE, IN, ER, RE, ON, AN, EN,..., and triple letter frequencies THE, AND, TIO, ATI, FOR, THA, TER, RES,... and so on.

➤ If a polyalphabetic cipher system was used then we will use the Kasiski method to find the length of the key k (number of alphabets). Then we divide the ciphertext into k parts, each part will analyze as in ❷ above.

The Kasiski method was introduced in 1863 by the Prussian military officer Friedrich W. Kasiski. The method analysis repetitions in the ciphertext to determine the period.

For example, consider the plaintext TO BE OR NOT TO BE enciphered with a Vigenere cipher with key HAM:

K=	H	A	M	H	A	M	H	A	M	H	A	M	H
M=	T	O	B	E	O	R	N	O	T	T	O	B	E
C=	A	O	N	L	O	D	U	O	F	A	O	N	L

The ciphertext contains two occurrences of the sequence AONL 9 characters apart, and the period could be 1,3 or 9 (we know it's 3).

Repetitions in the ciphertext more than two characters long are unlikely to occur by chance. They occur when the plaintext pattern repeats at a distance equal to a multiple of the period.

If there are m ciphertext repetitions that occur at intervals I_j ($1 \leq j \leq m$) the period is likely to be some number that divides most of the m intervals.

Example: We shall use IC and Kasiski method to analyze the following ciphertext.

ZHYME	ZVELK	OJUBW	CEYIN	CUSML	RAVSR	YARNH	CEARI	UJPGP	VARDU
QZCGR	NNCAW	JALUH	GJPJR	YGEGQ	FULUS	QFFPV	EYEDQ	<u>GOLKA</u>	<u>LVOSJ</u>
TFRTR	YEJZS	RVNCI	HYJNM	ZDCRO	DKHCR	MMLNR	FFLFN	<u>QGOLK</u>	<u>ALVOS</u>
<u>JWMIK</u>	QKUBP	SAYOJ	RRQYI	NRNYC	YQZSY	EDNCA	LEILX	RCHUG	IEBKO
YTHGV	VCKHC	<u>JEQGO</u>	<u>LKALV</u>	<u>OSJED</u>	WEAKS	GJHYC	LLFTY	IGSVT	FVPMZ
NRZOL	CYUZS	FKOQR	YRTAR	ZFGKI	QKRSV	IRCEY	USKVT	MKHCR	MYQIL
XRCRL	GQARZ	OLKHY	KSNFN	RRNCZ	TWUOC	JNMKC	MDEZP	IRJEJ	W

When we calculate the frequency distribution, we will find that the IC=0.04343, n=346,

$$k = \frac{0.0265 \cdot 346}{(0.065 - 0.04343) + 346(0.04343 - 0.03846)} = 5.2659$$

The IC indicates that this is a polyalphabetic cipher with a period of about 5.

We observe that there are 3 occurrences of the sequence QGOLKALVOSJ, the first two occurrences are separated by 51 and the last two by 72 characters (start to start); the only common divisor of 51 and 72 is 3 - the period is almost certainly 3.

Example: When we calculate the IC of some ciphertext, we find that k=9.34. Also we observe that there is NYX appearance many times in the ciphertext and the distance between them are 30, 50, 90, 110, and 33.

Since these can each be factored as

$$30 = 2 \times 3 \times 5$$

$$50 = 2 \times 5 \times 5$$

$$90 = 2 \times 3 \times 3 \times 5$$

$$110 = 2 \times 5 \times 11$$

$$33 = 3 \times 11$$

there are a number of candidates for key length. 2 and 5 are popular factors among these distance followed by 3 and 11. Note that all but 33 have $2 \times 5 = 10$ as a factor. The cryptanalyst might then disregard 33 as a pure coincidence, and discard that data in favor of conjecture that the key length is a multiple of 2 and/or 5. Combining this with data from the Friedman test that the key approximately 9 letters long, the cryptanalyst guesses that the key is 10 letters long, and not 2 or 5 letters long.

(1.8) Cryptanalysis:

The science of deducing the plaintext from a ciphertext, without knowledge of the key.

(1.8.1) Attacks on encrypted messages

The objective of the following attacks is to systematically recover plaintext from ciphertext, or even more drastically, to deduce the decryption key.

1. **A ciphertext-only attack:** is one where the adversary (or cryptanalyst) tries to deduce the decryption key or plaintext by only observing ciphertext. Any encryption scheme vulnerable to this type of attack is considered to be completely insecure.
2. **A known-plaintext attack:** is one where the adversary has a quantity of plaintext and corresponding ciphertext. This type of attack is typically only marginally more difficult to mount.
3. **A chosen-plaintext attack:** is one where the adversary chooses plaintext and is then given corresponding ciphertext. Subsequently, the adversary uses any information deduced in order to recover plaintext corresponding to previously unseen ciphertext.

4. **An adaptive chosen-plaintext attack:** is a chosen-plaintext attack wherein the choice of plaintext may depend on the ciphertext received from previous requests.
5. **A chosen-ciphertext attack:** is one where the adversary selects the ciphertext and is then given the corresponding plaintext. One way to mount such an attack is for the adversary to gain access to the equipment used for decryption (but not the decryption key, which may be securely embedded in the equipment). The objective is then to be able, without access to such equipment, to deduce the plaintext from (different) ciphertext.
6. **An adaptive chosen-ciphertext attack:** is a chosen-ciphertext attack where the choice of ciphertext may depend on the plaintext received from previous requests.

Some concepts on cryptanalysis:

- ✦ **Frequency:** number of appearance of the letter in the ciphertext, where the frequencies of the ciphertext letters are compared with the frequencies in Table 1 or Figure 5.
- ✦ **Repetition:** is the similar parts in the ciphertext that have length not less than three. This helps us to find the length of the key (the number of alphabets that used to enciphering in the polyalphabetic systems).
Take the Highest Common Factor HCF between the repetitions, which represent the length of the key, this method, is called the Kasiski method.

Letter	%	Letter	%
a	8.167	n	6.749
b	1.492	o	7.507
c	2.782	p	1.929
d	4.253	q	0.095
e	12.702	r	5.987
f	2.228	s	6.327
g	2.015	t	9.056
h	6.094	u	2.758
i	6.966	v	0.978
j	0.153	w	2.360
k	0.772	x	0.150
l	4.025	y	1.974
m	2.406	z	0.074

Table 1: English letters frequencies
English Letter Frequency

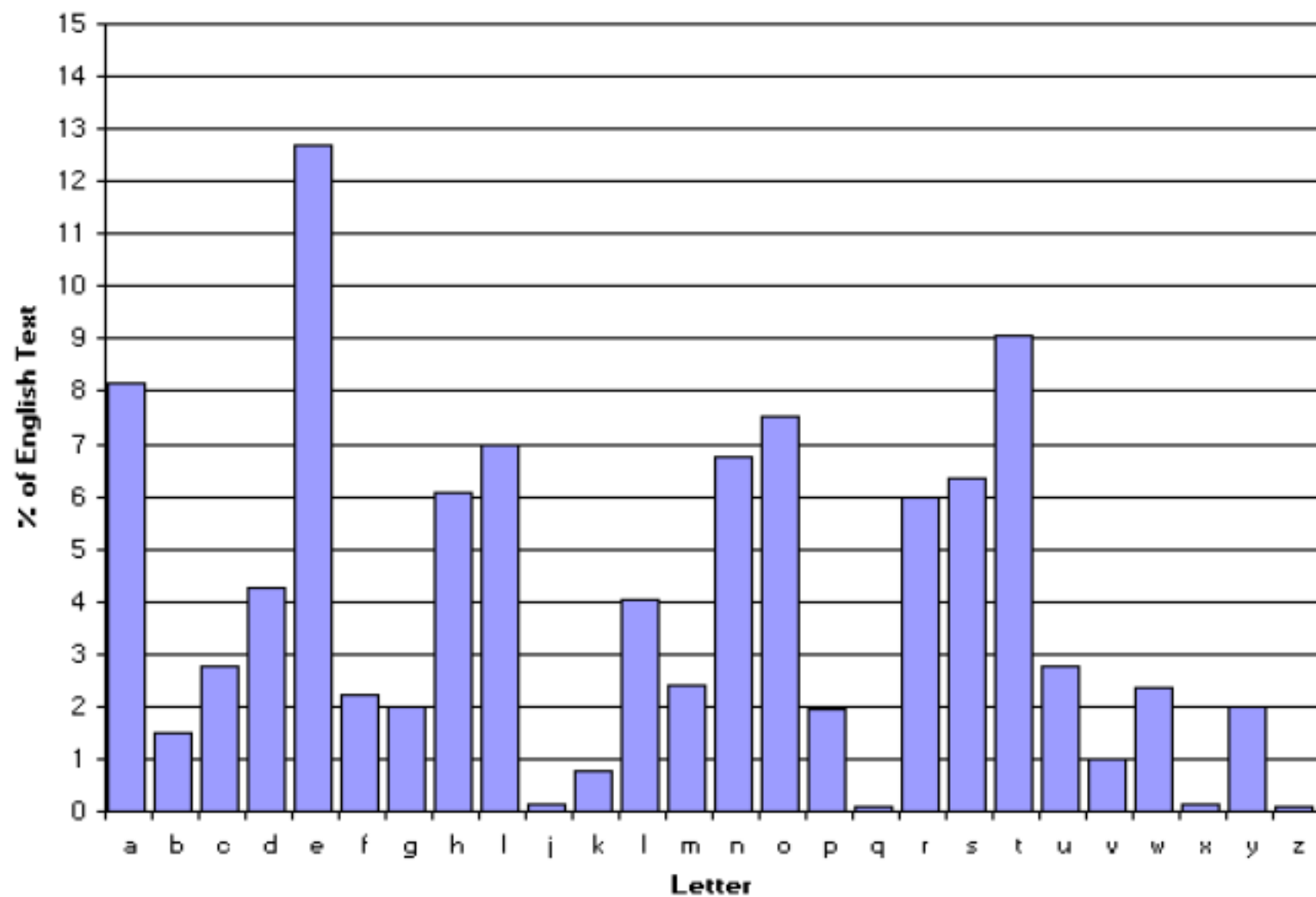


Figure 5: Histogram of English letters frequencies

- ✦ **Index of Coincidence (IC):** is the probability that two letters selected from the text are identical, we can compute the IC from the following equation:

$$IC = \frac{\sum_{\lambda} f_{\lambda}(f_{\lambda} - 1)}{n(n - 1)},$$

where f_{λ} is the frequency of the letter λ in the ciphertext and n is the length of the letter. The IC value differs from language to another. We can use the IC to discover if the message were enciphered using Monoalphabetic system or polyalphabetic system.

- ✦ **Coincidence:** is the computing of the coincidence of the ciphertexts, where two messages put one over the other, and the purpose is to discover if the two messages were enciphered using the same key. If there is 7 coincidence letters between 100 letters in the two messages then the two messages were enciphered using the same key, while if there is 4 letters coincidence between every 100 letters then they enciphered with different keys.

Cryptanalysis examples:

First of all we must specify the type of the cipher system that was used. If the frequencies of the ciphertext are the same as the frequencies of the language then, a transposition cipher system was used; otherwise a substitution cipher system was used.

1- Cryptanalysis of transposition cipher systems:

When we decide that a transposition cipher system were used, we put the cipher text in $m \times n$ matrix, m and n depends on the length of the received ciphertext, for example if the length is 500 then one of the possible sizes is 20×25 . Then we rearrange the columns to get some known patterns such as (and, the, ion, that,...) in addition to some expected word in the message.

As we know there are two types of transposition cipher system: simple and double transposition, the cryptanalysis of the last one is more complicated because we lose the ability to find the known patterns.

2- Cryptanalysis of substitution cipher systems:

If we know that a substitution cipher system was used, the next step is to determine whether a monoalphabetic system or polyalphabetic system was used, by using the IC of the language.

Example: A sample of ordinary English contains the following distribution of letters

Letter	Count	Letter	count
A	141	N	119
B	36	O	132
C	36	P	28
D	103	Q	1
E	188	R	95
F	37	S	64
G	34	T	182
H	102	U	59
I	123	V	13
J	4	W	55
K	18	X	3
L	56	Y	23
M	27	Z	0

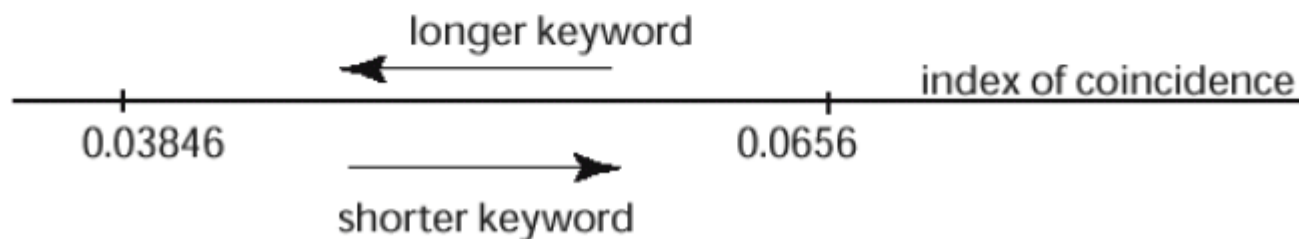
What is the probability of selecting an identical pair of letters from this collection? in other word compute the IC.

$$IC = \frac{\sum_A f_{\lambda}(f_{\lambda} - 1)}{n(n - 1)}$$

$$IC = \frac{141(141 - 1) + 36(36 - 1) + \dots + 23(23 - 1) + 0(0 - 1)}{1679(1679 - 1)} = \frac{184838}{2817362} \approx 0.0656.$$

As we see from the two examples above the index of coincidence of totally random (uniformly distributed) collection of letters is about 0.0385. Vigenere ciphertexts from longer keywords have a more uniform distribution of letters. For keyword length closer to 1, the index of coincidence will be larger, closer to 0.0656.

Polyalphabeticity Measure for English



If the length of the text is n , we can quantify the connection between index of coincidence and keyword length k , (number of alphabets), where:

$$k \approx \frac{0.0265 \cdot n}{(0.065 - IC) + n(IC - 0.0385)},$$

Example: A polyalphabetic ciphertext has the following letter counts.

Letter	Count	Letter	count
A	60	N	28
B	50	O	83
C	42	P	44
D	64	Q	69
E	51	R	13
F	63	S	29
G	19	T	66
H	48	U	87
I	56	V	63
J	67	W	19
K	23	X	43
L	45	Y	39
M	44	Z	67

Estimate the keyword length.

Solution: There are $n=1282$ letters.

$$IC = \frac{60 \cdot 59 + 50 \cdot 49 + \dots + 67 \cdot 66}{1282 \cdot 1281} = \frac{35761}{821121} \approx 0.04355.$$

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Based only on this evidence, a reasonably likely keyword length is 5.

➤ Now, after the above tests if we conclude that a monoalphabetic cipher system was used, then:

❶ If a direct standard or reversed system were used, we compare the frequencies of the ciphertext with the frequencies of the English language, start by putting E against the letter with the higher frequency in the ciphertext, then we put the other letters sequentially.

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➤ If a polyalphabetic cipher system was used then we will use the Kasiski method to find the length of the key k (number of alphabets). Then we divide the ciphertext into k parts, each part will analyze as in ❷ above.

2. Euclidean Algorithm:

If have two numbers c, q that $c = q * d + r$, then $GCD(c, q) = GCD(d, r)$

Ex1: find the Greatest Common Divisor (GCD) between 132 and 55 by using Euclid's Algorithm.

$$132 = 55 * 2 + 22$$

$$55 = 22 * 2 + 11$$

$$22 = 11 * 2 + 0$$

Stopping when getting zero 0 then GCD is 11:

$$GCD(132, 55) = GCD(55, 22) = GCD(22, 11) = GCD(11, 0) = 11$$

Ex2: find the GCD (252 , 198) by using Euclid's Algorithm.

$$252 = 198 * 1 + 54$$

$$198 = 54 * 3 + 36$$

$$54 = 36 * 1 + 18$$

$$36 = 18 * 2 + 0$$

$$GCD(252, 198) = (198, 54) = (54, 36) = (36, 18) = (18, 0) = 18$$

Public Key Cryptography and the RSA Algorithm

Private-Key Cryptography

- traditional **private/secret/single key** cryptography uses **one** key
- Key is shared by both sender and receiver
- if the key is disclosed communications are compromised
- also known as **symmetric**, both parties are equal
 - hence does not protect sender from receiver forging a message & claiming is sent by sender

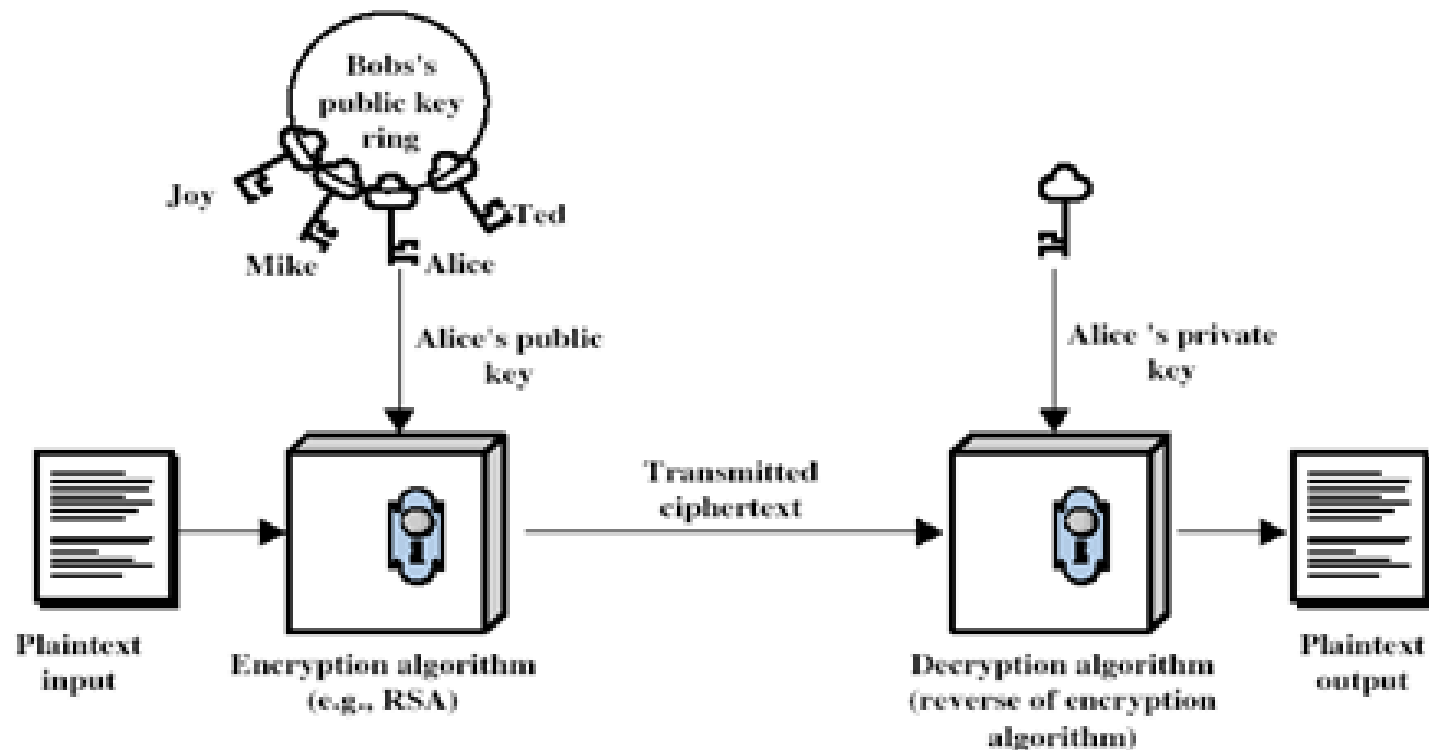
Public-Key Cryptography

- probably most significant advance in the 3000 year history of cryptography
- uses **two** keys – a public key and a private key
- **asymmetric** since parties are **not** equal
- uses clever application of number theory concepts to function
- complements **rather than** replaces private key cryptography

Public-Key Cryptography

- **public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
 - a **public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
 - a **private-key**, known only to the recipient, used to **decrypt messages**, and **sign** (create) **signatures**
- is **asymmetric** because
 - those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures

Public-Key Cryptography



Why Public-Key Cryptography?

- developed to address two key issues:
 - **key distribution** – how to have secure communications in general without having to trust a KDC with your key
 - **digital signatures** – how to verify a message comes intact from the claimed sender
- public invention due to Whitfield Diffie & Martin Hellman at Stanford U. in 1976
 - known earlier in classified community

Public-Key Characteristics

- Public-Key algorithms rely on two keys with the characteristics that it is:
 - computationally infeasible to find decryption key knowing only algorithm & encryption key
 - computationally easy to en/decrypt messages when the relevant (en/decrypt) key is known
 - either of the two related keys can be used for encryption, with the other used for decryption (in some schemes)

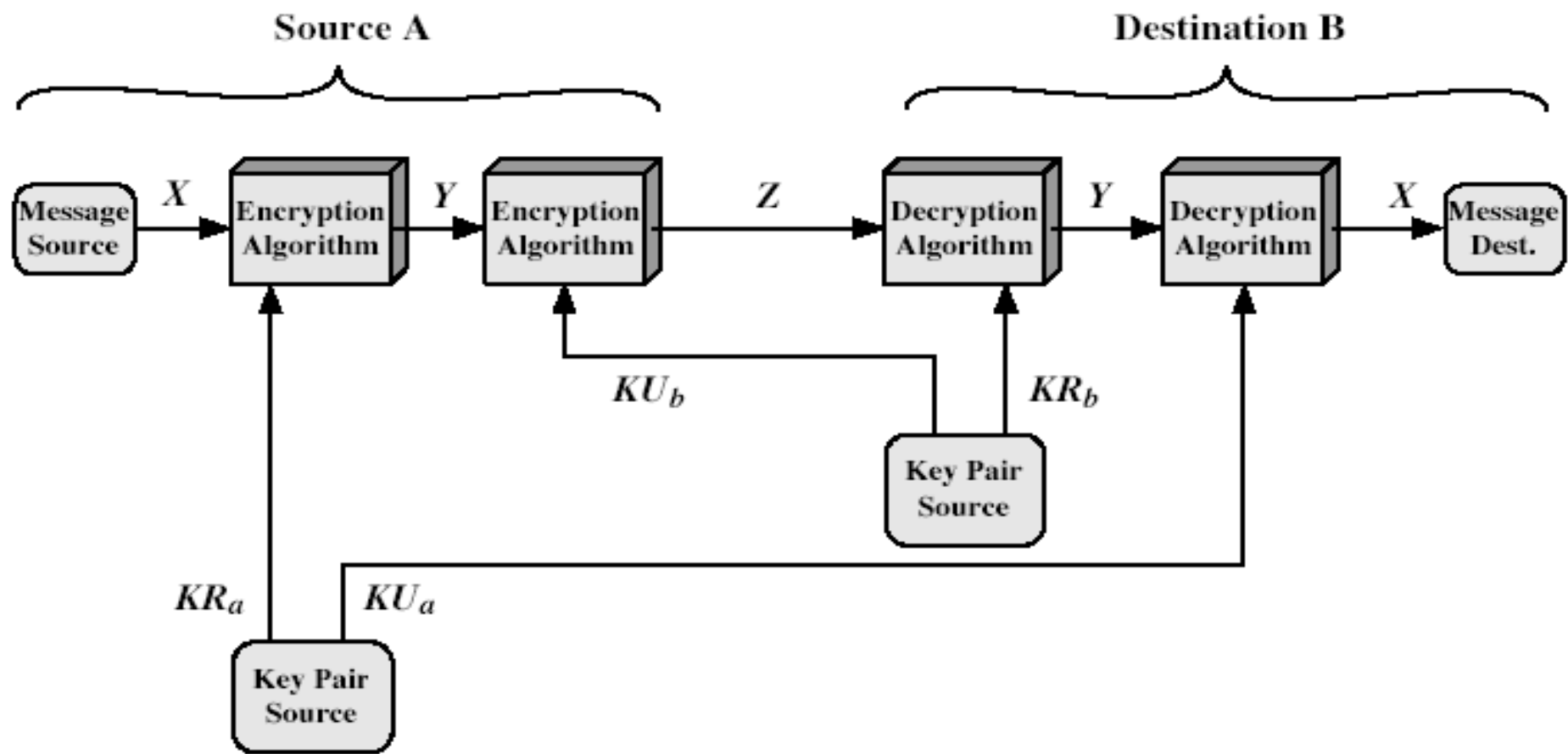


Figure 9.4 Public-Key Cryptosystem: Secrecy and Authentication

Public-Key Applications

- can classify uses into 3 categories:
 - **encryption/decryption** (provide secrecy)
 - **digital signatures** (provide authentication)
 - **key exchange** (of session keys)
- some algorithms are suitable for all uses, others are specific to one

Security of Public Key Schemes

- like private key schemes brute force **exhaustive search** attack is always theoretically possible
- but keys used are too large (>512 bits)
- security relies on a **large enough** difference in difficulty between **easy** (en/decrypt) and **hard** (cryptanalyse) problems
- more generally the **hard** problem is known, its just made too hard to do in practise
- requires the use of **very large numbers**
- hence is **slow** compared to private key schemes

RSA

- by Rivest, Shamir & Adleman of MIT in 1977
- best known & widely used public-key scheme
- based on exponentiation in a finite (Galois) field over integers modulo a prime
 - nb. exponentiation takes $O((\log n)^3)$ operations (easy)
- uses large integers (eg. 1024 bits)
- security due to cost of factoring large numbers
 - nb. factorization takes $O(e^{\log n \log \log n})$ operations (hard)

RSA Key Setup

- each user generates a public/private key pair by:
- selecting two large primes at random - p, q
- computing their system modulus $N=p \cdot q$
 - note $\phi(N) = (p-1)(q-1)$
- selecting at random the encryption key e
 - where $1 < e < \phi(N)$, $\gcd(e, \phi(N)) = 1$
- solve following equation to find decryption key d
 - $e \cdot d = 1 \bmod \phi(N)$ and $0 \leq d \leq N$
- publish their public encryption key: $KU = \{e, N\}$
- keep secret private decryption key: $KR = \{d, p, q\}$

RSA Use

- to encrypt a message M the sender:
 - obtains **public key** of recipient $KU = \{e, N\}$
 - computes: $C = M^e \bmod N$, where $0 \leq M < N$
- to decrypt the ciphertext C the owner:
 - uses their private key $KR = \{d, p, q\}$
 - computes: $M = C^d \bmod N$
- note that the message M must be smaller than the modulus N (block if needed)

Why RSA Works

- because of Euler's Theorem:
- $a^{\varphi(N)} \bmod N = 1$
 - where $\gcd(a, N) = 1$
- in RSA have:
 - $N = p \cdot q$
 - $\varphi(N) = (p-1)(q-1)$
 - carefully chosen e & d to be inverses mod $\varphi(N)$
 - hence $e \cdot d = 1 + k \cdot \varphi(N)$ for some k
- hence :
$$\begin{aligned} C^d &= (M^e)^d = M^{1+k \cdot \varphi(N)} = M^1 \cdot (M^{\varphi(N)})^k = M^1 \cdot (1)^k \\ &= M^1 = M \bmod N \end{aligned}$$

RSA Example

1. Select primes: $p=17$ & $q=11$
2. Compute $n = \underline{pq} = 17 \times 11 = 187$
3. Compute $\phi(n) = (p-1)(q-1) = 16 \times 10 = 160$
4. Select e : $\underline{\gcd}(e, 160) = 1$; choose $e=7$
5. Determine d : $de = 1 \pmod{160}$ and $d < 160$
Value is $d=23$ since $23 \times 7 = 161 = 10 \times 160 + 1$
6. Publish public key $KU = \{7, 187\}$
7. Keep secret private key $KR = \{23, 17, 11\}$

RSA Example cont

- sample RSA encryption/decryption is:
- given message $M = 88$ (nb. $88 < 187$)
- encryption:
$$C = 88^7 \bmod 187 = 11$$
- decryption:
$$M = 11^{23} \bmod 187 = 88$$

Exponentiation

- can use the Square and Multiply Algorithm
- a fast, efficient algorithm for exponentiation
- concept is based on repeatedly squaring base
- and multiplying in the ones that are needed to compute the result
- look at binary representation of exponent
- only takes $O(\log_2 n)$ multiples for number n
 - eg. $7^5 = 7^4 \cdot 7^1 = 3 \cdot 7 = 10 \pmod{11}$
 - eg. $3^{129} = 3^{128} \cdot 3^1 = 5 \cdot 3 = 4 \pmod{11}$

$c \leftarrow 0; d \leftarrow 1$

for $i \leftarrow k$ **downto** 0

do $c \leftarrow 2 \times c$

$d \leftarrow (d \times d) \bmod n$

if $b_i = 1$

then $c \leftarrow c + 1$

$d \leftarrow (d \times a) \bmod n$

return d